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Measurement of the tau lepton reconstruction and identification performance in the ATLAS experiment using pp collisions at $\sqrt{s} = 13$ TeV

The ATLAS Collaboration

This document details measurements of the performance of the reconstruction and identification of hadronic tau lepton decays using the ATLAS detector. The performance of these algorithms is measured with Z boson or top quark decays to tau leptons and uses the full 2015 dataset of pp collisions collected at the LHC, corresponding to an integrated luminosity of 3.2 fb^{-1} and a centre-of-mass energy $\sqrt{s} = 13$ TeV. The measurements include the performance of the identification, trigger, energy calibration, and electron discrimination algorithms for reconstructed tau candidates. The simulation to data correction factor on the offline tau identification efficiency is measured with a relative precision of 5% (6%) for one (three) track reconstructed tau candidates. For hadronic tau lepton decays selected by offline algorithms, the correction factor on the tau trigger identification efficiency is measured with a relative precision of 3–8% for $p_T(\tau_{\text{had}}) < 100$ GeV and 8–14% for $100 < p_T(\tau_{\text{had}}) < 300$ GeV, depending on the transverse energy and the number of associated tracks. The correction factor on the tau energy scale is measured with a relative precision of 2% (3%) for one (three) track reconstructed tau candidates. The correction factor on the probability of misidentifying an electron as a hadronic tau lepton decay is measured with a relative precision of between 3–14%, depending on the pseudorapidity of the reconstructed tau candidate.



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1 Introduction

With a mass of 1.777 GeV and a proper decay length of 87 μm [1], tau leptons decay either leptonically ($\tau \rightarrow \ell \nu_\ell \nu_\tau$, $\ell = e, \mu$) or hadronically ($\tau \rightarrow \text{hadrons } \nu_\tau$, labelled as τ_{had}) and do so typically before reaching active regions of the ATLAS detector. In this note, only hadronic tau lepton decays are considered. The hadronic tau lepton decays represent 65% of all possible decay modes. The hadronic decay products contain one or three charged pions in 72% and 22% of all cases, respectively. In 68% of all hadronic decays, at least one associated neutral pion is also produced. The neutral and charged hadrons stemming from the tau lepton decay make up the visible part of the tau lepton, and in the following are referred to as $\tau_{\text{had-vis}}$.

The main background to hadronic tau lepton decays is from jets of energetic hadrons produced via the fragmentation of quarks and gluons; Both are present at trigger level (also referred to as *online* in the following) as well as during the event reconstruction (referred to as *offline*). Discriminating variables based on the narrow shower in the calorimeter, the distinct number of tracks and the displaced tau lepton decay vertex are used to distinguish $\tau_{\text{had-vis}}$ candidates from jets. Electrons also form an important background for $\tau_{\text{had-vis}}$ containing one charged hadron. Therefore a specific method to suppress this background is devised too.

Final states with hadronic tau lepton decays are an important part of the ATLAS physics program. This places strong requirements on both $\tau_{\text{had-vis}}$ reconstruction and identification algorithms, as well as the performance measurements of the algorithms. The algorithms involved in triggering, reconstructing and identifying tau leptons during proton-proton collisions with a centre-of-mass energy $\sqrt{s} = 8$ TeV are described in Ref. [2], and the updates to these algorithms for $\sqrt{s} = 13$ TeV data collected in 2015 are described in Ref. [3].

This note first gives an overview of the tau reconstruction, energy calibration and identification algorithms and describes further updates to these algorithms for 2016 data-taking. The performance measurements of the triggering, energy calibration and identification algorithms using the 2015 data are described in the remaining sections. The performance of online and offline tau identification, and the tau energy scale calibration are measured using a *tag-and-probe* method applied to events enriched in the $Z \rightarrow \tau\tau$ process, with one tau lepton decaying to muon and neutrinos, τ_μ (tag), and the other decaying to hadrons and neutrino, τ_{had} (probe). The performance of the online and offline tau identification algorithms in simulation and in recorded data are measured and correction factors are derived. For the tau energy scale measurement, the reconstructed visible mass distribution of the muon and $\tau_{\text{had-vis}}$ system is determined in both data and simulation, and the energy calibration required to obtain agreement is calculated.

In order to extend the range of the p_T spectrum of tau candidates, the performance of the online tau identification algorithm is also measured using events enriched in the $t\bar{t}$ process. This measurement similarly uses the tag-and-probe method with a muon (tag) and a hadronic tau lepton decay (probe). Finally, the performance of the electron rejection algorithm is measured. The tag-and-probe method is used in events enriched in $Z \rightarrow ee$ decays featuring at least one electron (tag) and a tau candidate (probe), and the efficiency of the electron rejection algorithm is measured and correction factors are derived.

This note is organised as follows. After a description of the ATLAS detector in section 2, the data and simulation samples used in the studies presented are described in section 3. The reconstruction and requirements on the objects used in this note are described in section 4, with the tau reconstruction, energy calibration, identification, electron discrimination and trigger algorithms described separately in more detail in section 5. The tau identification, trigger and energy scale performance measurements using

the tag-and-probe method in $Z \rightarrow \tau_\mu \tau_{\text{had}}$ events are described in section 6. Similarly the tag-and-probe studies carried out using $t\bar{t}$ and $Z \rightarrow ee$ events are described in sections 7 and 8 respectively. Finally, a summary of the measurements is given and their use in physics analyses detailed.

2 ATLAS detector

The ATLAS detector [4] consists of an inner tracking system surrounded by a superconducting solenoid, electromagnetic (EM) and hadronic (HAD) calorimeters, and a muon spectrometer (MS).

The inner detector is immersed in a 2 T axial magnetic field, and consists of silicon pixel and microstrip (SCT) detectors inside a transition radiation tracker (TRT), providing charged particle tracking in the region $|\eta| < 2.5$ ¹. For the $\sqrt{s} = 13$ TeV run, a fourth layer of the pixel detector, the Insertable B -Layer (IBL) [5], has been inserted at an average radius of 33.2 mm, providing an additional position measurement with 8 μm resolution in the (x, y) plane and 40 μm along z .

The EM calorimeter uses lead and liquid argon (LAr) as absorber and active materials, respectively. In the central rapidity region, the EM calorimeter is divided in three layers, one of them segmented in thin η strips for optimal γ/π^0 separation, completed by a presampler layer for $|\eta| < 1.8$. Hadron calorimetry is based on different detector technologies, with scintillator tiles ($|\eta| < 1.7$) or LAr ($1.5 < |\eta| < 4.9$) as active media, and uses steel, copper, or tungsten as the absorber material. The calorimeters provide coverage within $|\eta| < 4.9$. The MS consists of superconducting air-core toroids, a system of trigger chambers covering the range $|\eta| < 2.4$, and high-precision tracking chambers allowing muon momentum measurements within $|\eta| < 2.7$.

The ATLAS trigger system consists of two levels which reduce the initial bunch crossing rate to a manageable rate for disk storage while keeping interesting physics events. The first level (L1) is hardware-based and uses a subset of the detector information to reduce the accepted event rate to 100 kHz [6]. This is followed by a software-based High Level Trigger (HLT) that further reduces the average recorded collision rate to around 1 kHz.

3 Data and simulation samples

The data used in this note were recorded by the ATLAS experiment during the 2015 LHC run with proton-proton collisions at a centre-of-mass energy of $\sqrt{s} = 13$ TeV. They correspond to an integrated luminosity of 3.2 fb^{-1} . To ensure good data quality, the inner-detector tracking systems, calorimeters and muon spectrometer are required to be fully operational.

The simulated background processes consist of the production of Z +jets, W +jets, single top quarks and $t\bar{t}$ pairs. They are modelled with several event generators as described below, while contributions from multi-jet production are estimated with data-driven techniques as described in each section.

¹ ATLAS uses a right-handed coordinate system with its origin at the nominal interaction point (IP) in the centre of the detector and the z -axis along the beam direction. The x -axis points from the IP to the centre of the LHC ring, and the y -axis points upward. Cylindrical coordinates (r, ϕ) are used in the transverse (x, y) plane, ϕ being the azimuthal angle around the beam direction. The pseudorapidity is defined in terms of the polar angle θ as $\eta = -\ln \tan(\theta/2)$. The distance ΔR in the $\eta - \phi$ space is defined as $\Delta R = \sqrt{(\Delta\eta)^2 + (\Delta\phi)^2}$.

Simulated samples of Z+jets events and W+jets events are produced using POWHEG-BOX v2 [7–9] interfaced to PYTHIA 8.186 [10] with the AZNLO tune [11]. In this sample, PHOTOS++ v3.52 [12, 13] is used for photon radiation from electroweak vertices and charged leptons. All W/Z+jets samples use the CT10 PDF set and are normalised to the next-to-next-to-leading-order (NNLO) cross sections calculated using FEWZ [14, 15].

The POWHEG-BOX v2 program with the CT10 PDF set is used for the generation of $t\bar{t}$ pairs and single top quarks in the Wt - and s -channels. Samples of t -channel single-top-quark events are produced with the POWHEG-BOX v1 [7, 8] generator employing the four-flavour scheme for the NLO matrix element calculations together with the fixed four-flavour scheme PDF set CT10f4; the top-quark decay is simulated with MadSpin [16]. For all samples of top-quark production, the spin correlations are preserved and the parton shower, fragmentation and underlying event are simulated using PYTHIA 6.428 [17] with the CTQ6L1 PDF set and the corresponding Perugia 2012 tune [18]. Photon radiation from charged leptons and electroweak vertices is simulated using PHOTOS++ v3.52. The top-quark mass is set to 172.5 GeV. The $t\bar{t}$ production sample is normalised to the NNLO cross section, including soft-gluon resummation to next-to-next-to-leading-logarithm accuracy (Ref. [19] and references therein). The normalisation of the single top quark event samples uses an approximate NNLO calculation from Refs. [20–22].

The effect of multiple proton (pp) interactions, referred to as pile-up, is simulated by overlaying minimum-bias interactions on the generated events. The simulated events are reweighted such that the average number of pp interactions per bunch crossing has the same distribution in data and simulation.

4 Object selection

Muons are reconstructed by combining an inner detector track with a track from the MS [23], and must lie within $|\eta| < 2.5$, with a transverse momentum requirement of $p_T > 10$ GeV for the $Z \rightarrow ee$ analysis and $p_T > 7$ GeV for all other analyses. Corrections to simulated reconstruction efficiencies, derived from the data, are applied to the simulated samples. The muon candidates have to pass a “loose” muon identification requirement, which corresponds to a 98% efficiency.

Electrons are reconstructed by matching clustered energy deposits in the electromagnetic calorimeter to tracks reconstructed in the inner detector, and are required to have $p_T > 15$ GeV and $|\eta| < 2.47$ (excluding the transition region between the barrel and end-cap calorimeters, corresponding to the region $1.37 < |\eta| < 1.52$). Electrons are identified using the signal and background probability density functions for several discriminating variables detailed in Ref. [24, 25]. A likelihood (LLH) function is constructed with several working points available, corresponding to cuts on the likelihood score with different levels of signal efficiency and background rejection. For the studies in this document, the *medium* identification working point is required to select electron candidates and the *very loose* working point is used for vetoing electrons faking taus. Corrections to the reconstruction and identification efficiencies derived from the data are applied to the simulated samples.

For muons and electrons, the scalar sum of the transverse momenta of tracks within a cone of p_T -dependent size, $\Delta R < \min(10 \text{ GeV}/p_T, 0.3)$, centred on the lepton candidate track and excluding the lepton track, is required to be less than a p_T -dependent fraction of the lepton transverse momentum. Additionally, the sum of the calorimeter energy deposits in a cone of size $\Delta R < 0.2$ around the lepton, excluding energy associated with the lepton candidate, must be less than a p_T dependent percentage of the lepton energy.

This requirement, referred to as *gradient isolation*, is used in the electron rejection and trigger efficiency measurements, and has a 90 (99)% efficiency at 25 (60) GeV.

Another isolation criterion uses a similar definition, except with a fixed cone size of $\Delta R < 0.4$ for tracks and with the threshold values fixed at 1% and 4% for the sum of track momenta, and the sum of the calorimeter energy deposits respectively. This isolation, referred to as *fixed-threshold* isolation, provides a stronger multi-jet rejection. Fixed-threshold isolation is used to select electrons and muons in the offline tau identification and tau energy scale measurements.

Jets are constructed using the anti- k_t algorithm [26], with a distance parameter $R = 0.4$. Three-dimensional clusters of calorimeter cells called *TopoClusters* [27], calibrated using a local hadronic calibration (LC) [28], serve as inputs to the jet algorithm. Jets are required to be within $|\eta| < 4.5$. A dedicated b -tagging algorithm described in Ref. [29, 30] is used to identify jets associated with the decay of a b -quark in the range $|\eta| < 2.5$, and has a 77% efficiency.

The reconstruction, energy calibration and identification of the hadronic decays of tau leptons are described in the next section whilst the geometric overlap of objects with $\Delta R < 0.2$ is resolved by selecting only one of the overlapping objects in the following order of priority: muons, electrons, $\tau_{\text{had-vis}}$ candidates, and jets. The missing transverse momentum, with magnitude E_T^{miss} , is calculated from the vector sum of the transverse momenta of all reconstructed electrons, muons, $\tau_{\text{had-vis}}$ and jets in the event, as well as a term for the remaining tracks [31].

5 Reconstruction, energy calibration and identification of hadronic tau decays

This section gives an overview of the online and offline algorithms used in the reconstruction, energy calibration and identification of the hadronic decays of tau leptons. A detailed description of the offline algorithms used in the reconstruction, baseline energy calibration and identification of hadronic tau decays during the 2015 data-taking period of Run 2, as well as the expected performance of the reconstruction algorithms and associated systematic uncertainties can be found in Ref. [3].

5.1 Tau reconstruction

Tau candidates are seeded by jets formed using the procedure described in Section 4. Jets seeding tau candidates are additionally required to have $p_T > 10$ GeV and $|\eta| < 2.5$. Tau candidates in the transition region between the barrel and forward calorimeters, $1.37 < |\eta| < 1.52$, are vetoed.

A tau vertex is chosen as the candidate track vertex with the largest fraction of momentum from tracks associated ($\Delta R < 0.2$) with the jet. The tracks must pass requirements on the number of hits in the tracker and have $p_T > 1$ GeV. Additional requirements are placed on the shortest distance from the track to the tau vertex in the transverse plane, $|d_0| < 1$ mm, and the shortest distance in the longitudinal plane, $|\Delta z_0 \sin(\theta)| < 1.5$ mm where θ is the polar angle of the track and z_0 is the point of closest approach along the longitudinal axis. Provided these requirements are passed the tracks are then associated to *core* ($0 < \Delta R < 0.2$) and *isolation* ($0.2 < \Delta R < 0.4$) regions around the tau candidate.

The direction (eta/phi) of the tau candidate is calculated using the vectorial sum of the TopoClusters within $\Delta R < 0.2$ of the seed jet barycenter, using the tau vertex as the origin. The mass of the tau candidate

is defined to be zero and consequently the transverse momentum, p_T , and the transverse energy, E_T are identical. The energy of the tau is obtained through dedicated calibration schemes, described in the following section.

5.2 Tau energy calibration

A tau-specific energy calibration is applied to the tau candidate in order to correct the energy deposition measured in the detector to the average value of the energy carried by the measured decay products at the generator level. Two calibrations are available, known as the baseline calibration, and the boosted regression tree (BRT) based calibration. With the exception of the BRT-specific tau energy scale measurement presented in section 6.4, all measurements performed in this note use the baseline calibrated tau candidates.

The baseline correction to the tau energy is calculated as:

$$E_{\text{calib}} = \frac{E_{\text{LC}} - E_{\text{pileup}}}{\mathcal{R}(E_{\text{LC}} - E_{\text{pileup}}, |\eta|, n_p)} . \quad (1)$$

First, a correction to the LC-calibrated sum of the energy of all TopoClusters within $\Delta R < 0.2$ of the tau candidate, E_{LC} , is applied for the energy contribution, $E_{\text{pile-up}}$, from multiple interactions occurring in the same bunch crossing (referred to as pileup). It is found that $E_{\text{pile-up}}$ increases linearly with the number of primary vertices N_{PV} . Thus, the pile-up correction is derived as:

$$E_{\text{pileup}}(N_{\text{PV}}, |\eta|, n_p) = A(|\eta|, n_p) \times (N_{\text{PV}} - \langle N_{\text{PV}} \rangle), \quad (2)$$

and calculated separately for 1-track and 3-track tau candidates, denoted by n_p . Following the Run 1 convention, the pile-up correction, $A(|\eta|, n_p) \times (N_{\text{PV}} - \langle N_{\text{PV}} \rangle)$, vanishes for a value of N_{PV} equal to the average pile-up conditions of the MC samples, e.g. $N_{\text{PV}} \approx 14$. The linear coefficient A is extracted from a linear fit of E_{LC} from simulated samples as a function of $(N_{\text{PV}} - \langle N_{\text{PV}} \rangle)$ in bins of $|\eta|$. The detector response calibration, $\mathcal{R}$, extracted as the Gaussian mean of the $(E_{\text{LC}} - E_{\text{pileup}}) / E_{\text{true}}^{\text{vis}}$ distribution, is then applied. Here, $E_{\text{true}}^{\text{vis}}$ is the energy of the generated tau decay products, including final state radiation, but excluding the contribution of the neutrinos

The energy resolution of the baseline method is excellent at high p_T but quickly degrades at low p_T . A method of reconstructing the individual charged and neutral hadrons in tau decays has recently been developed by the ATLAS experiment and is known as "Tau Particle Flow" (TPF) [32]. The method significantly improves the tau energy resolution at low p_T due to the superior measurement of the charged pion momentum from the tracking system. An improved energy calibration has therefore been introduced which combines the information from the baseline and TPF methods together with additional calorimeter and tracking information via a multivariate-analysis technique. This technique is referred to as a BRT method, and is implemented using the TMVA package [33].

To train the BRT, tau candidates satisfying the medium tau identification requirement, which is described in Section 5.3, coming from simulated $Z/\gamma^* \rightarrow \tau\tau$ events are used. The two figures of merit, used to determine which input variables and BRT tuning parameters are optimal, are defined as follows: the *resolution* is defined as half of the 68% central interval of the ratio of the calibrated $\tau_{\text{had-vis}}$ transverse momentum to the generated $\tau_{\text{had-vis}}^{\text{true,vis}}$ transverse momentum, $p_T^{\text{true,vis}}$, whilst the *non-closure* is the offset of the most probable value of that ratio from unity. The most probable value is used instead of the arithmetic

mean in this case to reduce the impact of the asymmetric tails of the distribution. The transverse component of the sum of the momenta of the reconstructed charged hadron and neutral pion constituents is referred to as p_T^{TPF} , and the transverse momentum at LC-scale is p_T^{LC} . As the resolution of p_T^{TPF} is better than p_T^{LC} at low p_T , and vice-versa at higher p_T , the interpolated transverse momentum, p_T^{interp} , is defined in the following equation:

$$p_T^{\text{interp}} = f_x \times p_T^{\text{LC}} + (1 - f_x) \times p_T^{\text{TPF}}, \quad (3)$$

where f_x is a weight between zero and one and is a function of p_T^{LC} :

$$f_x(p_T^{\text{LC}}) = \frac{1}{2} \left(1 + \tanh \frac{p_T^{\text{LC}} - x \text{ GeV}}{20 \text{ GeV}} \right). \quad (4)$$

The symbol x defines the point where the transition from low p_T to high p_T occurs, and is chosen to be $x = 250$. The regression target of the BRT training is the ratio of the generated $\tau_{\text{had-vis}}$ transverse momentum to p_T^{interp} .

The final input variables used in the BRT are listed and described in Table 1. The transverse momenta p_T^{LC} and p_T^{TPF} provide basic knowledge about the $\tau_{\text{had-vis}}$ energy. The BRT is less powerful when two variables are highly correlated and so to reduce the correlation, ratios of these variables $p_T^{\text{LC}}/p_T^{\text{interp}}$ and $p_T^{\text{TPF}}/p_T^{\text{interp}}$ are used instead of the raw values.

The moments of the individual TopoCluster constituents of the tau candidates, such as λ_{centre} , $\langle \lambda^2 \rangle$, $\langle \rho \rangle$, $f_{\text{presampler}}$, and P_{EM} , used in the LC calibration as described in Ref. [28], are found to be powerful inputs to the BRT-based tau energy calibration. In order to simplify the use of these variables, they are re-defined at the per-tau level, as the energy-weighted average of these moments for all constituent TopoClusters. The variables μ and n_{PV} are included to provide information about multiple interactions occurring in the same bunch crossing, whilst γ_π and n_{π^0} are variables that provide information about the decay modes of the tau candidate and improve the resolution at low p_T .

Figure 1 shows the $\tau_{\text{had-vis}}$ energy resolution of baseline, TPF-based and BRT-based calibrations. In the region $p_T < 100$ GeV, the BRT-based tau energy calibration improves on the baseline resolution by a factor of two, while at high p_T the performance is comparable.

5.3 Tau identification

The tau identification algorithm is designed to reject backgrounds from quark- and gluon-initiated jets. The identification uses Boosted Decision Tree (BDT) based methods [34, 35]. As described in Ref. [3], the BDT for tau candidates associated with one and three tracks are trained separately with simulated $Z/\gamma^* \rightarrow \tau\tau$ for signal and dijet events (selected from data) for background. Three working points labelled *loose*, *medium* and *tight* are provided, and correspond to different tau identification efficiency values, with the efficiency designed to be independent of p_T . The target efficiencies are 0.6, 0.55 and 0.45 for the generated 1-track loose, medium and tight working points, and 0.5, 0.4 and 0.3 for the corresponding generated 3-track target efficiencies. The input variables to the BDT are corrected such that the mean of their distribution for signal samples is constant as a function of pile-up. This ensures that the efficiency for each working point does not depend strongly on the pile-up conditions. The figures of all the input variables can be found in Appendix A

Number of primary vertices, n_{PV}
Number of primary vertices in the event.
Average interactions per crossing, μ
Average number of interactions per bunch crossing.
Cluster shower depth, λ_{centre}
Distance of the cluster shower centre from the calorimeter front face measured along the shower axis.
Cluster second moment in λ, $\langle \lambda^2 \rangle$
Second moment of the distance of a cell, λ , from the shower centre along the shower axis.
Cluster first moment in energy density, $\langle \rho \rangle$
Cluster first moment in energy density $\rho = E/V$, where E and V represent the energy and volume of the cluster, respectively.
Cluster presampler fraction, $f_{\text{presampler}}$
Fraction of cluster energy deposited in the barrel and endcap presamplers.
Cluster EM-like probability, P_{EM}
Classification probability of the cluster to be EM-like, as described in Ref. [28].
Number of associated tracks, n_{track}
Number of tracks associated with the $\tau_{\text{had-vis}}$.
Number of reconstructed neutral pions, n_{π^0}
Number of reconstructed neutral pions associated with the $\tau_{\text{had-vis}}$.
Relative difference of pion energies, γ_{π}
Relative difference of the total charged pion energy, E_{charged} , and the total neutral pion energy, E_{neutral} : $\gamma_{\pi} = (E_{\text{charged}} - E_{\text{neutral}})/(E_{\text{charged}} + E_{\text{neutral}})$.
Calorimeter-based pseudorapidity, η_{calo}
Calorimeter-based (baseline) pseudorapidity.
Interpolated transverse momentum, $p_{\text{T}}^{\text{interp}}$
Transverse momentum interpolated from calorimetric corrections to energy measurement and TPF reconstruction.
Ratio of p_{T}^{LC} to $p_{\text{T}}^{\text{interp}}$, $p_{\text{T}}^{\text{LC}}/p_{\text{T}}^{\text{interp}}$
Ratio of the local hadron calibration transverse momentum to $p_{\text{T}}^{\text{interp}}$.
Ratio of $p_{\text{T}}^{\text{TPF}}$ to $p_{\text{T}}^{\text{interp}}$, $p_{\text{T}}^{\text{TPF}}/p_{\text{T}}^{\text{interp}}$
Ratio of the TPF reconstruction transverse momentum, $p_{\text{T}}^{\text{TPF}}$, to $p_{\text{T}}^{\text{interp}}$.

Table 1: List of input variables used in the boosted regression tree for $\tau_{\text{had-vis}}$ energy calibration. The cluster variables are defined as the energy-weighted average of the moments for all constituents TopoClusters, as described in detail in Ref. [28].

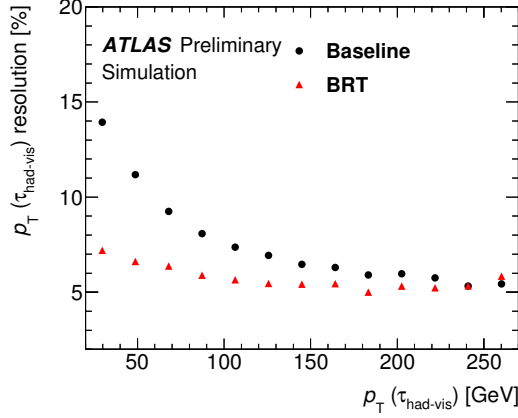


Figure 1: The resolution of the baseline and the BRT-based $\tau_{\text{had-vis}}$ energy calibration. The resolution, shown as a function of the generated tau p_T , is defined as half of the 68% central intervals of the ratio of the calibrated p_T to the true visible p_T .

5.4 Electron discrimination

In order to reduce the electron background, reconstructed tau candidates associated with a single track and within a distance of $\Delta R < 0.4$ of a reconstructed electron are rejected if the electron passes a *very loose* working point of the electron likelihood identification algorithm, as mentioned in Section 4. This electron likelihood score cut is tuned to yield a 95% efficiency for hadronically decaying taus, and the cut on the electron identification likelihood score is dependent on the p_T and $|\eta|$ of the tau candidate. The cut value on the electron identification likelihood has been updated in comparison to Ref. [3] to reflect the 2015 data-taking conditions [25].

5.5 Tau trigger identification

Due to the technical limitations of the ATLAS trigger system [36], the online reconstruction of hadronic tau lepton decays is modified compared to the offline reconstruction. At L1, trigger towers are defined in the EM and hadronic calorimeters with a granularity of $\Delta\eta \times \Delta\phi = 0.1 \times 0.1$. A core region is made up of a set of 2×2 trigger towers and a requirement is placed on the transverse energy sum of the two most energetic adjacent EM calorimeter towers. A requirement can also be placed on the transverse energy deposited in an isolation region of calorimeter towers around the core.

At HLT, the energy is recalculated using LC-calibrated TopoClusters of calorimeter cells contained in a $\Delta R = 0.2$ cone around the L1 tau direction. The energy is calibrated using the baseline method described in Section 5.2, using dedicated factors derived to account for the different LC calibration constants and pile-up correction scheme used at trigger level. Additionally, the pile-up subtraction term in the baseline calibration is calculated using the average number of interactions per crossing instead of the number of primary vertices, as the latter is not available at trigger level. A minimum transverse energy requirement is placed on the online tau candidate.

A multi-stage tracking algorithm is then used to reconstruct tracks in a sufficiently large ΔR cone around the tau candidate. The first stage, referred to as fast-tracking, reconstructs tracks using trigger-specific

pattern recognition algorithms in two steps. In the first step the highest- p_T track is identified in a narrow cone ($\Delta R < 0.1$) around the tau direction and along the direction of the beam in the range $|z| < 255$ mm. The second step reruns the fast-tracking software in an extended cone ($\Delta R < 0.4$) around the tau direction but with the tracks required to emanate from the same position along the beamline as the leading track. A requirement is made on the number of tracks within a core ($\Delta R < 0.2$) and isolation ($0.2 < \Delta R < 0.4$) region of $0 < N_{\text{core}}^{\text{tracks}} < 4$ and $N_{\text{iso}}^{\text{tracks}} < 2$, respectively. Finally, the detector hits and tracks identified in the fast-tracking stages are used as seeds in an HLT-tracking stage. HLT-tracking refers to the precision reconstruction of tracks with algorithms similar to those used in the offline reconstruction.

The HLT precision track and calorimeter information is used to calculate a number of pileup corrected variables which are then input into an online tau BDT. The variables used in the online BDT are the same as those used in the offline BDT, with the exception that the lifetime variables are calculated with respect to the beamspot instead of the tau vertex, as full event vertex reconstruction is not available at the HLT. The offline BDT training is used directly at trigger level, as it was found to provide good rejection while minimizing the orthogonality between the trigger and offline selection. As in the case of offline tau identification, three identification working points are available for online tau candidates with one or three associated tracks: loose, medium and tight. These working points were tuned to provide target efficiencies of approximately 0.95 (0.70) after their offline counterpart identification is applied for taus with one (three) associated tracks

6 $Z \rightarrow \tau_\mu \tau_{\text{had}}$ tag-and-probe analyses

In this section, the performance of the identification, trigger and energy reconstruction algorithms are evaluated with a data sample enriched in $Z \rightarrow \tau_\mu \tau_{\text{had}}$ events where one tau lepton decays to a muon and the other decays hadronically, with associated neutrinos. The chosen tag-and-probe approach consists of selecting events triggered by the presence of a muon (tag) and containing a hadronically decaying tau lepton candidate (probe) in the final state and studying the performance of the identification, trigger and energy reconstruction algorithms. In this section *signal* refers to a $\tau_{\text{had-vis}}$ candidate geometrically matched with a generated $\tau_{\text{had-vis}}$ or the $Z \rightarrow \tau_\mu \tau_{\text{had}}$ event containing such tau candidates.

6.1 Common event selection

To select $Z \rightarrow \tau_\mu \tau_{\text{had}}$ events, a single-muon trigger with the threshold of $p_T > 20$ GeV is used. The offline reconstructed muon candidate must have $p_T > 22$ GeV and be geometrically matched to the online muon. Events are required to have no additional electrons or muons and at least one $\tau_{\text{had-vis}}$ candidate with 1 or 3 tracks. If there are multiple $\tau_{\text{had-vis}}$ candidates, only the leading p_T candidate is considered. In addition, a very loose requirement on the tau identification BDT output (> 0.3) is made which suppresses jets while being more than 99% efficient for the generated $\tau_{\text{had-vis}}$. The muon and $\tau_{\text{had-vis}}$ candidates are required to have opposite-sign electric charges (OS). Events with b -tagged jets are rejected, after which the contribution from top quark backgrounds is reduced from 1-3% to 0.3-0.8%, depending on the analysis and the number of charged tracks associated to the tau candidate. The associated b -tagging systematic uncertainty is found to be negligible, since the probability of passing the b -jet veto is very high ($> 99\%$) for jets that do not originate from a b -quark.

A series of selection requirements is used to suppress W +jets (mainly $W \rightarrow \mu\nu_\mu$) events. The transverse mass of the muon and E_T^{miss} system, $m_T = \sqrt{2p_T^\mu \cdot E_T^{\text{miss}}(1 - \cos \Delta\phi(\mu, E_T^{\text{miss}}))}$, is required to be less than 50 GeV, where p_T^μ is the transverse momentum of the muon, and $\Delta\phi(\mu, E_T^{\text{miss}})$ is the $\Delta\phi$ separation between the muon and the missing transverse momentum. The quantity $\Sigma \cos \Delta\phi = \cos \Delta\phi(\mu, E_T^{\text{miss}}) + \cos \Delta\phi(\tau_{\text{had-vis}}, E_T^{\text{miss}})$ is required to be greater than -0.5 , where $\Delta\phi(\tau_{\text{had-vis}}, E_T^{\text{miss}})$ is the $\Delta\phi$ separation between the $\tau_{\text{had-vis}}$ and the missing transverse momentum.

In addition to the above common selection, the medium offline tau identification requirement is applied in the energy scale measurement. Several offline working points, i.e. loose, medium and tight, are applied in the online tau identification efficiency to derive the corresponding trigger efficiencies. A requirement on the invariant mass of the muon and tau candidate, $45 \text{ GeV} < m_{\text{vis}}(\mu, \tau_{\text{had-vis}}) < 80 \text{ GeV}$, is applied in both online and offline tau identification efficiency measurements, but not in the energy scale measurement since the $m_{\text{vis}}(\mu, \tau_{\text{had-vis}})$ distribution is used to constrain the tau energy scale. To reduce the large contamination from misidentified jets in the offline tau identification efficiency measurement, in which the medium tau identification requirement is not applied, the lower threshold on $\Sigma \cos \Delta\phi$ is tightened to -0.1 . The detailed event selections and the signal purity, i.e. the estimated fraction of generated tau leptons to the total predicted number of events after applying the signal selection requirements listed above, are summarised in Table 2.

Analyses	Offline Identification	Online Identification	TES
m_T	$< 50 \text{ GeV}$	$< 50 \text{ GeV}$	$< 50 \text{ GeV}$
$\Sigma \cos \Delta\phi$	> -0.1	> -0.5	> -0.5
m_{vis}	$(45\text{--}80 \text{ GeV})$	$(45\text{--}80 \text{ GeV})$	–
Tau Identification	–	various	medium
Purity	20% (before ID)	73% (medium)	65%

Table 2: Summary of the $Z \rightarrow \tau_\mu \tau_{\text{had}}$ event selections and purities in the online and offline tau identification, as well as the tau energy scale measurement.

After the final selection, besides a small fraction of muons misidentified as hadronic tau lepton decays (which are modelled via simulation), the main background for the probe $\tau_{\text{had-vis}}$ candidates are jets misidentified as hadronic tau lepton decays from W +jets and multi-jet events. In these jet to τ fake events, especially multi-jet events, the correlation of the charge sign between misidentified jets and the muon is not as strong as in the case of $Z \rightarrow \tau_\mu \tau_{\text{had}}$ signal events. Therefore, the events with same sign (SS) charge are used to model the jet to τ fake background. Due to the residual charge correlation, there are more background events with opposite sign charge than with same sign. To take this charge correlation into account, a normalisation factor on multi-jet events is measured from the multi-jet control region, defined below, by comparing events with opposite sign charge and same sign charge. For the W +jets events, the trigger and energy scale measurements use simulated events to model the difference between opposite sign events and same sign events, the so called ‘OS–SS’ method, with the correction factors derived from W +jets control region events with opposite sign charge and the same sign charge requirements, respectively. The offline identification measurement follows the same principle but takes a slightly different approach, as discussed in section 6.2.1.

To improve the modelling of the jet background, two control regions enriched in specific background processes are used. A W +jets control region, as shown in Figure 2, is selected by requiring $E_T^{\text{miss}} > 30 \text{ GeV}$ and $m_T > 60 \text{ GeV}$, and a multi-jet control region, as shown in Figure 3, is selected by inverting the muon

isolation requirement. The identification is applied in the control regions for the energy scale and the trigger measurements, while the offline identification measurement has the control regions both with and without the tau identification requirement, in order to extract the yield in both cases.

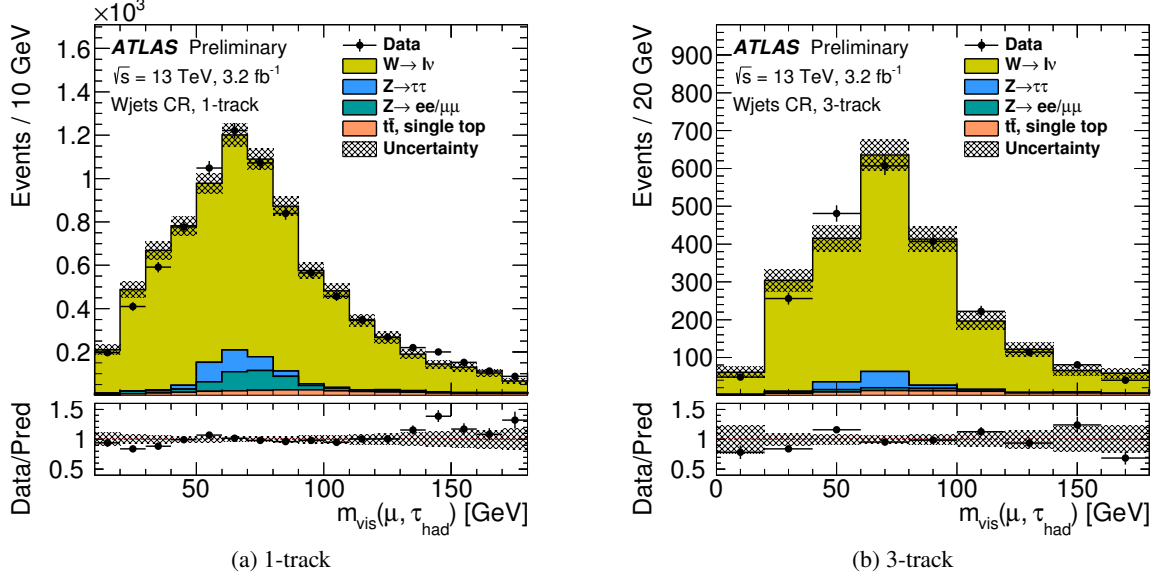


Figure 2: The distribution of m_{vis} , the invariant mass of the $\tau_{\text{had-vis}}$ and muon system, in the W +jet control region. The tau candidate is required to pass medium identification. The error band contains only the statistical uncertainty.

6.2 Offline tau identification efficiency measurement

In the offline tau identification efficiency measurement, the contamination from jet backgrounds before the application of the tau identification poses the greatest challenge. To estimate the background contamination in data, a template fit is performed using a variable with high separation between signal and background, to estimate the normalisation factors for both signal and background. The signal contribution to data before the application of the tau identification requirement is estimated directly from the fit, whilst the signal contribution to data after the application of the various tau identification working points is extracted by subtracting the estimated backgrounds from the data, after applying the same background normalisation factors as those before the identification is applied.

The variable used is the track multiplicity, defined as the sum of the number of core ($\Delta R < 0.2$) and outer ($0.2 < \Delta R < 0.6$) tracks associated to the $\tau_{\text{had-vis}}$ candidate. Outer tracks are only considered if they fulfil the track separation requirement, $D^{\text{outer}} = \min([p_{\text{T}}^{\text{core}}/p_{\text{T}}^{\text{outer}}] \cdot \Delta R(\text{core}, \text{outer})) < 4$, where $p_{\text{T}}^{\text{core}}$ refers to any track in the core region, and $\Delta R(\text{core}, \text{outer})$ refers to the distance between the candidate outer track and any track in the core region. More details can be found in Section 4 of Ref. [2].

6.2.1 Signal and background estimation

This section describes how the templates for the signal and the background are constructed and how the tau identification efficiency is measured. The tau candidate track multiplicity distribution of signal events

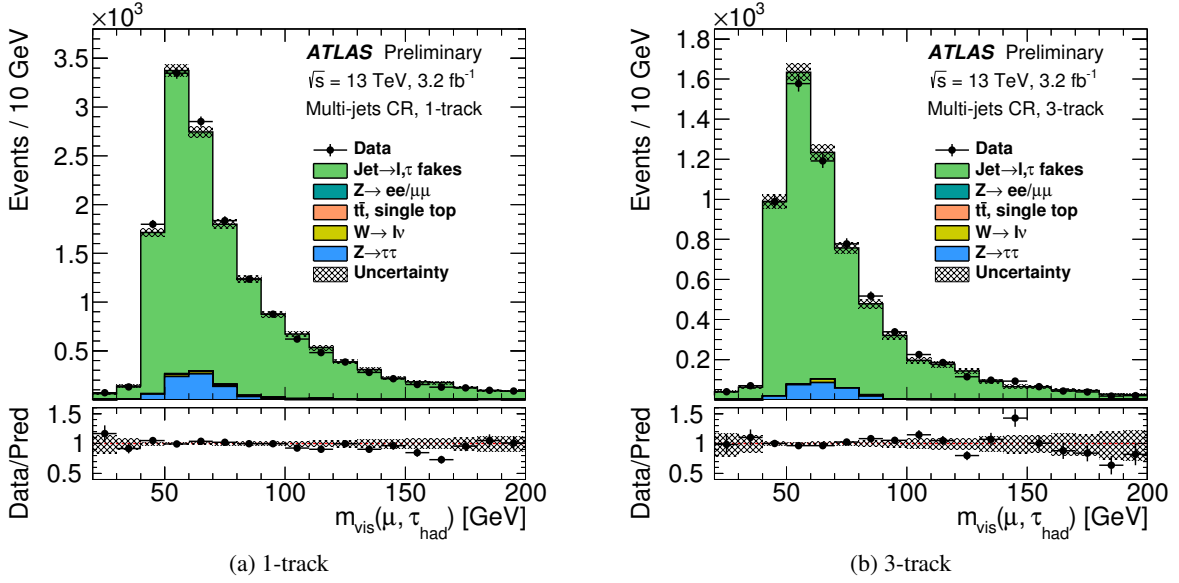


Figure 3: The distribution of m_{vis} , the invariant mass of the $\tau_{\text{had-vis}}$ and muon system, in the multi-jet control region. The tau candidate is required to pass medium identification. The error band contains only the statistical uncertainty.

is modelled using simulated $Z \rightarrow \tau_\mu \tau_{\text{had}}$ events with a small contribution from $t\bar{t}$ events. Two signal templates are obtained by requiring exactly one or three tracks reconstructed in the core region of the $\tau_{\text{had-vis}}$ candidate. To improve the fit stability, a combined template is used to model both the one track $\tau_{\text{had-vis}}$ and the three track $\tau_{\text{had-vis}}$ which are shown in separated colors in Figure 4. A single template is used to model the background from quark- and gluon-initiated jets that are misidentified as hadronic tau lepton decays:

$$F_{OS}^{bkg} = r_{QCD} \cdot (F_{SS}^{data} - F_{SS}^{W+jets CR} \cdot T_{SS}^{W+jets MC}) + F_{OS}^{W+jets CR} \cdot T_{OS}^{W+jets MC}, \quad (5)$$

where F terms represent the template in data and the W +jets control region for OS and SS, and T terms represent transfer factors, which are measured as the ratio of the simulated events with the signal region selection to simulated events with the W +jets control region selection for the OS and SS requirements, as a function of the track multiplicity. The background is mainly composed of multi-jet and W +jets events with a minor contribution from Z +jets events. Jet to tau fakes in $t\bar{t}$ events is included in the analysis and covered together as W +jets. The jet template is constructed starting with data events in the same sign control region (F_{SS}^{data}), a region enriched in events with jets misidentified as tau candidates. The contributions from W +jets and Z +jets in the SS control region ($F_{SS}^{W+jets CR} \cdot T_{SS}^{W+jets MC}$) are subtracted to yield the multi-jet only contribution. Then, to correct for the difference in normalisation between the same sign multi-jet events and the opposite sign ones, the template is scaled by the ratio (r_{QCD}) of OS/SS multi-jet events. To reduce the $Z \rightarrow \tau\tau$ signal contamination in the multi-jet control region, events with $45 \text{ GeV} < m_{\text{vis}}(\ell, \tau_{\text{had-vis}}) < 80 \text{ GeV}$ in the multi-jet control region are rejected. Finally, the OS contributions from W +jets events are added to complete the template. The shape of the W +jets contribution is estimated from the W +jets control region ($F_{OS}^{W+jets CR}$) and normalised to the signal region using transfer factors ($T_{OS}^{W+jets MC}$) derived using simulated W +jets events. The jet templates both with and without the identification requirements are built with this procedure.

An additional background is the contamination due to the misidentification of muons and electrons as tau candidates, and comes mainly from $Z \rightarrow \mu\mu$ processes. This small background contribution is modelled by simulated $Z \rightarrow ee/\mu\mu$, $t\bar{t}$, $Z \rightarrow \tau\tau$, diboson events where the reconstructed tau candidate probe is matched to a generated muon or electron. The yield of events where the lepton is misidentified as a tau candidate is typically less than 1% of the total signal yield.

With the normalisation factor of the signal and jet templates allowed to float, the signal plus background model of the track multiplicity is fitted to the data without identification. The normalisation factor of the jet template obtained from this fit is applied to the jet templates both before and after each tau identification working point is applied. The yield of the τ signal both before and after the application of each working point is obtained from the data subtracted by the estimated backgrounds from simulated muons and the jet template with the normalisation factor applied.

The efficiency for a given identification working point is calculated by taking the ratio of the extracted number of signal events before and after the identification criteria is applied.

6.2.2 Results

Figure 4 shows the track multiplicity distributions before and after applying the medium tau identification requirement. To account for the small differences between data and simulation, correction factors (also referred to as scale factors), defined as the ratio of the efficiency in data ($\varepsilon_{\text{Data}}$) to the efficiency in simulation (ε_{MC}) for $\tau_{\text{had-vis}}$ signal to pass a certain level of identification, are derived. The results are shown in Figure 5 and found to be compatible with unity.

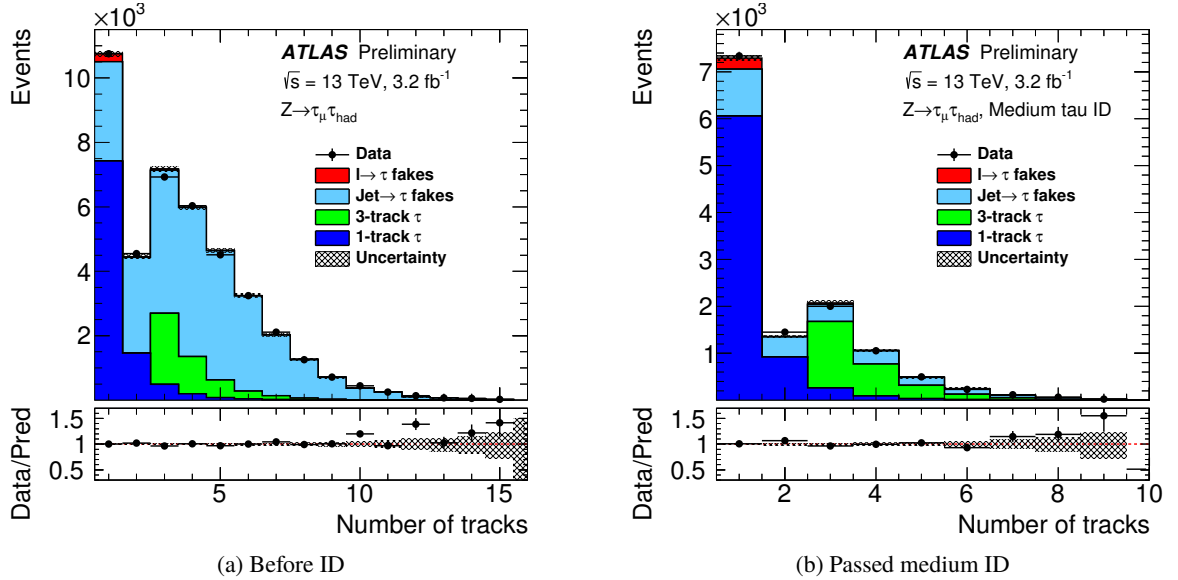


Figure 4: Track multiplicity: the sum of the number of core tracks and the outer tracks in $0.2 < \Delta R < 0.6$ that fulfil the requirement $D^{\text{outer}} < 4$, as defined in the text and Ref. [2]. The true tau and ‘Jet $\rightarrow$ τ fakes’ component are fitted to data while ‘lepton (l) $\rightarrow$ τ fakes’ component is fixed to the simulation prediction. The uncertainty band contains only the statistical uncertainty. Note: the track multiplicity included in the fit is up to 15, as shown in (a).

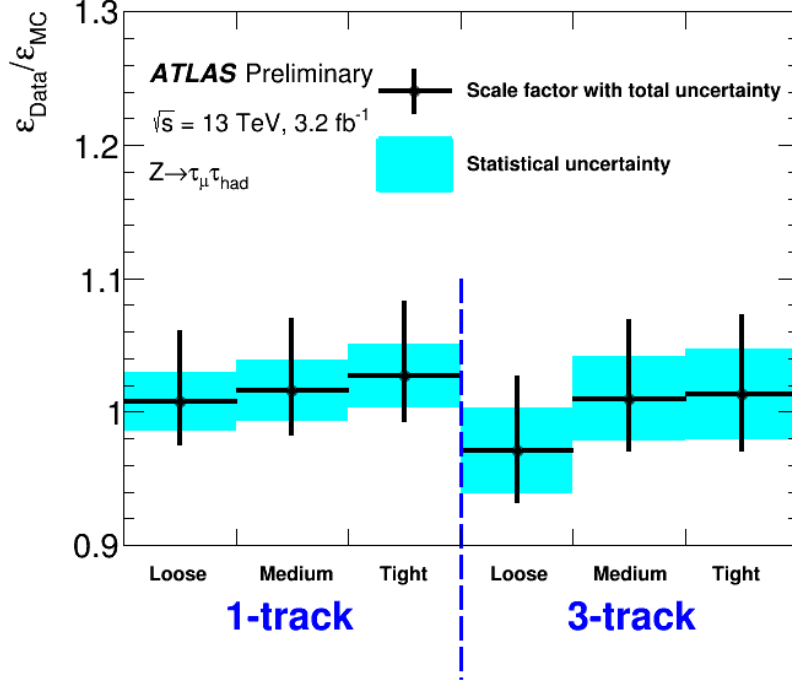


Figure 5: The scale factors ($\epsilon_{\text{Data}}/\epsilon_{\text{MC}}$) needed to bring the offline tau identification efficiency in simulation (ϵ_{MC}) to the level observed in data (ϵ_{Data}) for one track and three track $\tau_{\text{had-vis}}$ candidates with $p_T > 20$ GeV. The combined systematic and statistical uncertainties are shown.

The sources of uncertainty on the scale factors are summarised in Table 3. The uncertainty on the signal template is estimated by comparing simulated signal generated with different configurations, such as variations on the amount of detector material, and the hadronic interaction model, e.g. QGSP_BIC and FTFP_BERT models [37–40]. These comparisons are used to evaluate the effect of different simulation models on the relative normalisation of the 1-track to 3-track templates which are fixed to the simulated values in the fit. The uncertainty on the jet template accounts for differences between the W +jets shape in the signal and control regions and is derived from comparisons to simulated W +jets events, as well as the differences between the multi-jet template shape in the opposite sign and same sign region derived by varying the multi-jet control region selections. The uncertainty due to the normalisation of the misidentified muon background is estimated conservatively by varying the normalisation up and down by 50%.

Figure 6 shows the jet BDT score distribution, while Figures 7 to 10 show some of the input variables to the jet discriminant BDT. The figures of all the input variables can be found in Appendix A whilst the definition of the input variables can be found in Section 5 of Ref. [3]. For both the jet BDT score figure and the input variable figures, the distributions for the signal and background are obtained using the same methods as for the efficiency measurement. Some of the input variables comparisons show discrepancies; these are globally accounted for by the scale factors derived in the efficiency measurements.

Uncertainty source	Uncertainty [%]	
	1-track	3-track
Jet template modelling	1.5	1.5
Tau template modelling	4.4	4.3
Lepton template modelling	1.7	1.7
Statistics	1.7	2.8
Total	5.2	5.6

Table 3: Dominant uncertainties on the tau identification efficiency scale factors estimated with the Z boson tag-and-probe method, and the total uncertainty, which combines systematic and statistical uncertainties. These uncertainties apply to $\tau_{\text{had-vis}}$ candidates passing the medium tau identification algorithm with $p_T > 20$ GeV. The uncertainties for other tau identification working points are similar.

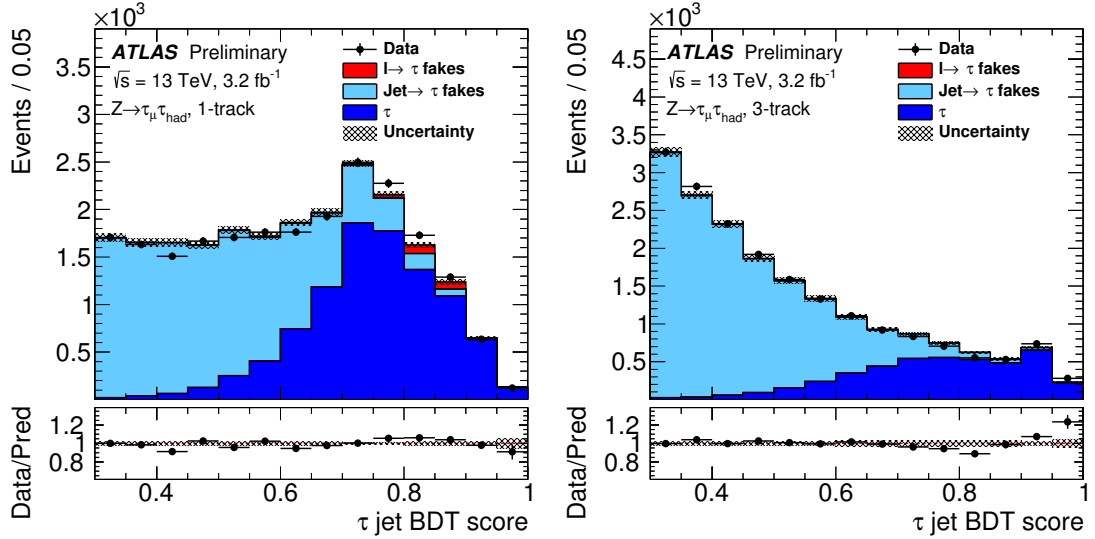


Figure 6: The jet discriminant BDT output distribution for one track (left) and three track (right) $\tau_{\text{had-vis}}$ candidates. The uncertainty band contains only the statistical uncertainty.

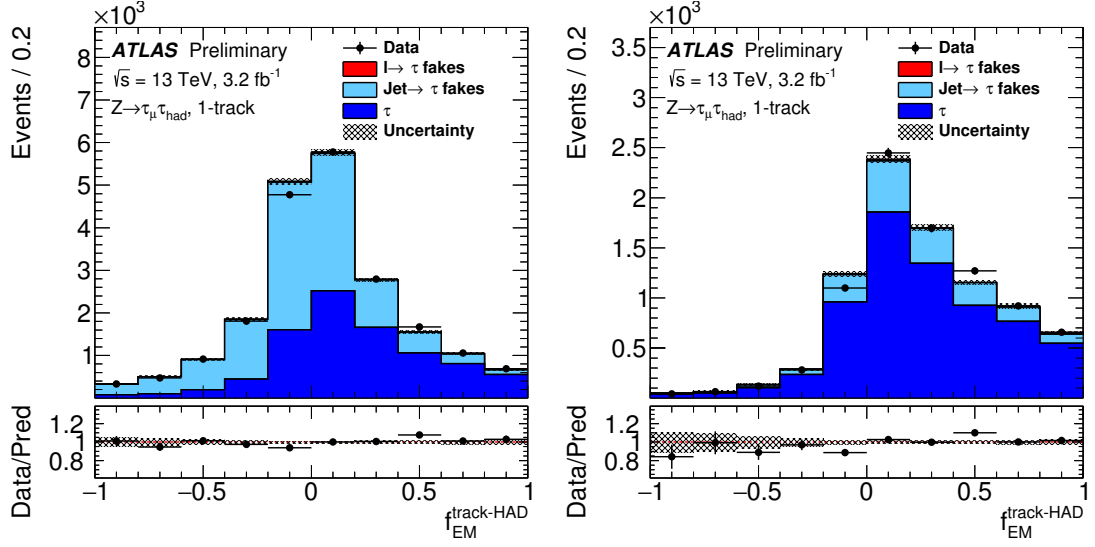


Figure 7: Fraction of EM energy from charged pions $f_{\text{EM}}^{\text{track-HAD}}$ for one track $\tau_{\text{had-vis}}$ candidates before (left) and after (right) the medium identification requirement. $f_{\text{EM}}^{\text{track-HAD}}$ is defined as the fraction of the energy, deposited in the electromagnetic calorimeter, of tracks associated with the $\tau_{\text{had-vis}}$ candidate in the core region. The uncertainty band contains only the statistical uncertainty.

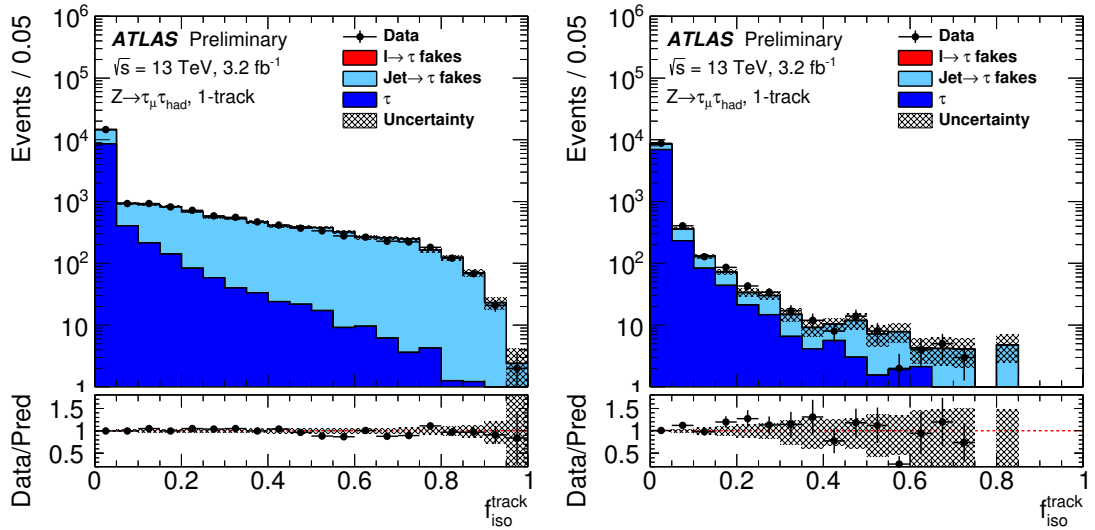


Figure 8: Fraction of tracks p_T in the isolation region $f_{\text{iso}}^{\text{track}}$ for one track $\tau_{\text{had-vis}}$ candidates before (left) and after (right) the medium identification requirement. $f_{\text{iso}}^{\text{track}}$ is defined as the scalar sum of the p_T of tracks associated with the $\tau_{\text{had-vis}}$ candidate in the region $0.2 < \Delta R < 0.4$ divided by the sum of the p_T of all tracks associated with the $\tau_{\text{had-vis}}$ candidate. The uncertainty band contains only the statistical uncertainty.

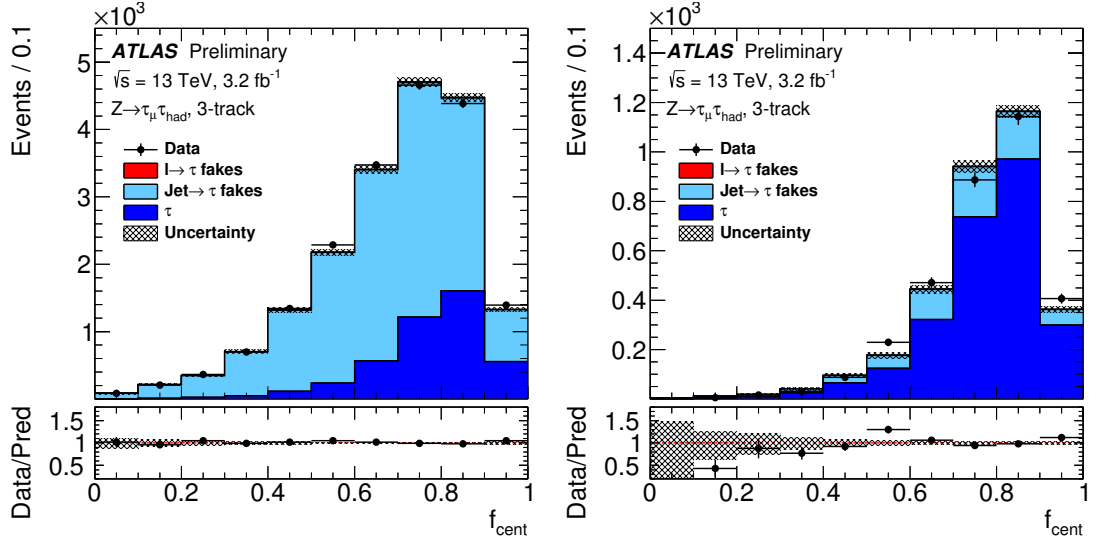


Figure 9: Central energy fraction f_{cent} for three track $\tau_{\text{had-vis}}$ candidates before (left) and after (right) the medium identification requirement. f_{cent} is defined as the fraction of the calorimeter transverse energy deposited in the region $\Delta R < 0.1$ with respect to all energy deposited in the region $\Delta R < 0.2$ around the $\tau_{\text{had-vis}}$ candidate. The uncertainty band contains only the statistical uncertainty.

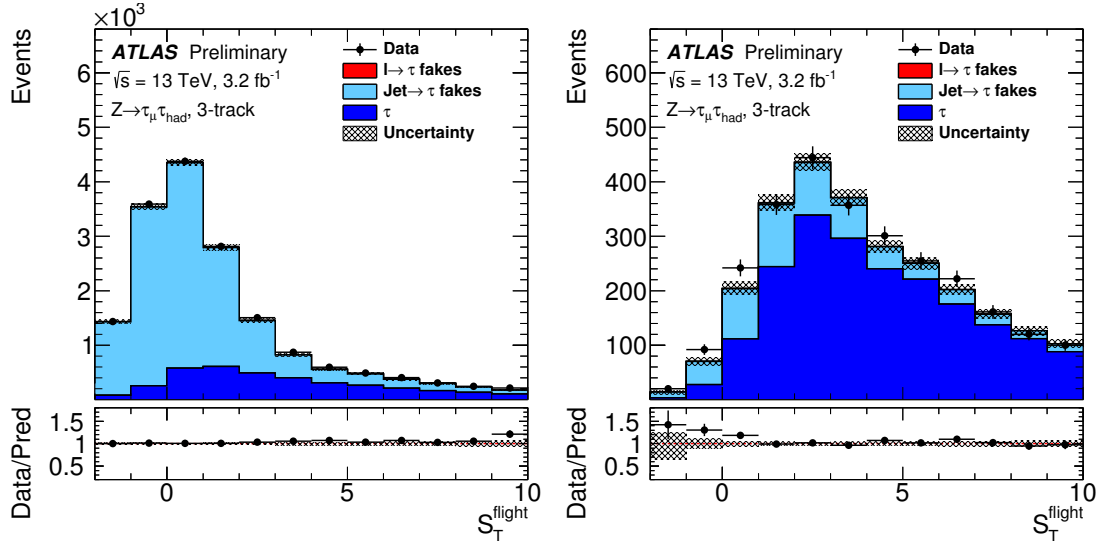


Figure 10: Transverse flight path significance S_T^{flight} for three track $\tau_{\text{had-vis}}$ candidates before (left) and after (right) the medium identification requirement. S_T^{flight} is defined as the decay length of the secondary vertex, reconstructed from the tracks associated with the core region of the $\tau_{\text{had-vis}}$ candidate, in the transverse plane, divided by its estimated uncertainty. The uncertainty band contains only the statistical uncertainty.

6.3 Trigger efficiency measurement

This section describes the trigger efficiency measurement in data using the tag-and-probe method, after the corresponding offline identification criteria is applied. The selection of $Z \rightarrow \tau_\mu \tau_{\text{had}}$ events used in this measurement is described in section 6.1.

6.3.1 Signal and background estimation

The dominant background contributions come from the misidentification of jets as $\tau_{\text{had-vis}}$ candidates in multi-jet and W +jets events which are estimated via data-driven methods using dedicated control regions. The multi-jet background is fully taken from the same sign control region with a normalisation factor (r_{QCD}) used to correct the difference between the same charge event and opposite charge events. The same sign component of other backgrounds, estimated from simulation, is subtracted to avoid double counting. The r_{QCD} normalisation factor is derived in the multi-jet control region as the ratio of the opposite charge yield to same charge yield.

The overall background in the signal region is given by:

$$N_{OS}^{bkg} = r_{\text{QCD}} \cdot N_{SS}^{data} + N_{MC}^{Z \rightarrow \tau\tau} + N_{MC}^{Z \rightarrow \ell\ell} + N_{MC}^{W+jets} + N_{MC}^{top}, \quad (6)$$

$$\text{with } N_{MC}^X = N_{MC, OS}^X - r_{\text{QCD}} \cdot N_{MC, SS}^X, \quad (7)$$

where, N_{SS}^{data} is the number of data events from the same sign control region, and $N_{MC, OS}^X$ ($N_{MC, SS}^X$) represents the number of simulated events for a given background, X , with the opposite (same) sign charge. To correct for the discrepancy between data and simulation in the W +jets background, a normalisation factor, k_{OS} (k_{SS}), is defined as the ratio of data to simulated W +jets events in the W +jets control region with the opposite (same) sign charge requirement. The number of W +jets events is thus given by:

$$N_{MC}^{W+jets} = k_{OS} \cdot N_{MC, OS}^{W+jets} - r_{\text{QCD}} \cdot k_{SS} \cdot N_{MC, SS}^{W+jets}. \quad (8)$$

The r_{QCD} and k_{OS} (k_{SS}) normalisation factors are derived separately for high p_T (> 40 GeV) and low p_T (< 40 GeV) tau candidates, as well as 1-track and 3-track tau candidates.

Figure 11 shows the signal region offline tau candidate p_T distributions before and after the application of the tau trigger.

6.3.2 Results

The tau trigger efficiency is measured with respect to tau candidates reconstructed and identified offline as a function of both the reconstructed transverse momentum of the tau candidate, and the number of reconstructed event vertices in the event. At L1, the tau trigger has requirements of $p_T > 12$ GeV and calorimetric isolation, whereby energy deposited in a ring surrounding the L1 tau object is required to be lower than a threshold dependent on the L1 tau energy. At HLT, the trigger has requirements of $p_T > 25$ GeV, the number of associated charged tracks restricted to three or less, and a medium working point selection on the online BDT score. Figure 12 shows the tau trigger efficiency measured in data (with estimated backgrounds subtracted) for the different levels of the trigger.

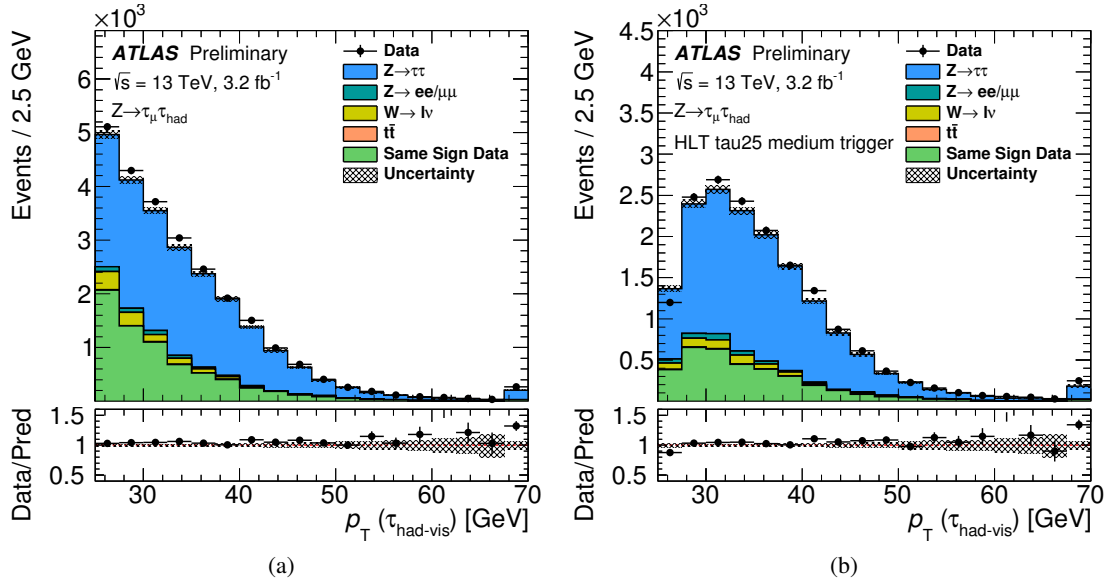


Figure 11: Offline p_T distribution of the tau candidate passing the offline medium tau identification (a) before, and (b) after the application of the tau trigger. The tau trigger has an online p_T requirement of 25 GeV, and a medium online identification. The events are from a selection designed to be enriched in the process $Z \rightarrow \tau_\mu \tau_{\text{had}}$. The uncertainty band contains only the statistical uncertainty.

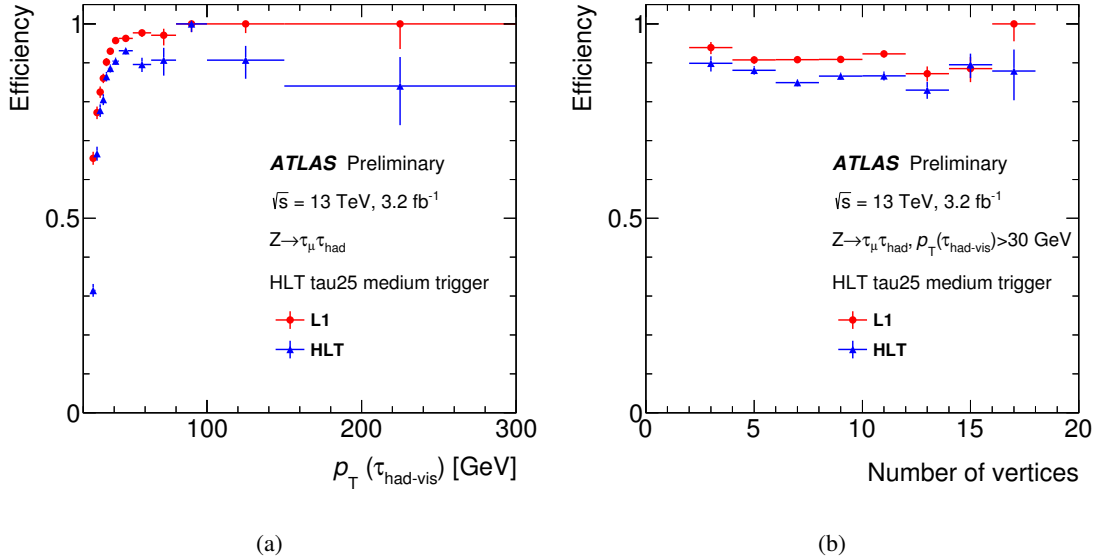


Figure 12: The Level 1 (red dots) and High Level Trigger (blue triangles) online tau identification efficiency for $\tau_{\text{had-vis}}$ candidates identified by the offline medium tau identification, as a function of (a) the offline $\tau_{\text{had-vis}}$ transverse momentum and (b) the number of primary vertices. The error bars correspond to the statistical uncertainty.

The measured online tau identification efficiency is compared to simulated $Z \rightarrow \tau\tau$ events in figure 13 and is shown to be well modelled outside the turn-on region. As in the offline tau identification efficiency measurement, scale factors are derived to account for the differences between data and simulation and are found to be consistent with unity for tau candidates with a reconstructed transverse momentum above 30 GeV.

The dominant systematic uncertainties considered for the efficiency measurement are shown in table 4, and are associated with the background subtraction. The largest systematic uncertainty results from the uncertainty on the estimation of the multi-jet background and contains a statistical component related to the number of events in the SS control region, and a systematic component derived by varying the choice of selection criteria used to define the control region. The uncertainty on the normalisation factor covers the uncertainty in the assumption that the OS to SS ratio is the same between the isolation and anti-isolation regions. The systematic uncertainties are larger in the low- p_T region due to the larger background contribution.

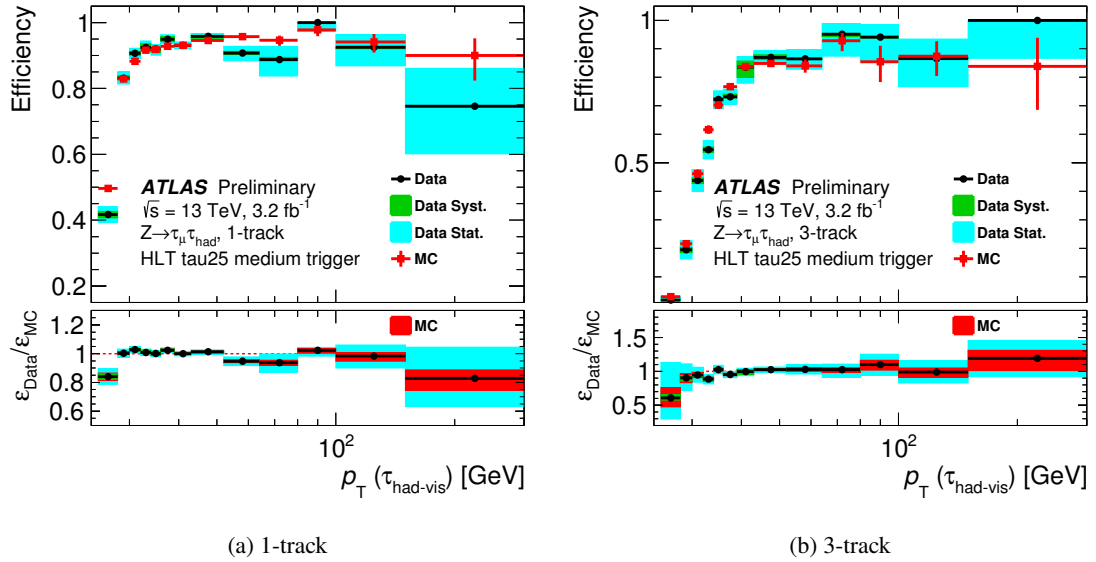


Figure 13: Efficiencies for signal and background subtracted data as well as the corresponding scale factors ($\epsilon_{\text{Data}}/\epsilon_{\text{MC}}$) for the tau trigger with online $p_T > 25$ GeV, as a function of the transverse momentum of offline $\tau_{\text{had-vis}}$ candidates with a medium identification requirement. “Data Syst.” (“Data Stat.”) reflects the systematic (statistical) uncertainty associated with data; “MC” includes only the statistical uncertainty of the simulation.

Uncertainty sources	1-track		3-track	
	Without trigger	With trigger	Without trigger	With trigger
Statistical	1.9%	2.5%	3.5%	7.8%
Jet fake background	3.0%	3.0%	7.3%	7.9%
Muon performance	0.5%	0.6%	0.8%	1.7%
Pileup and soft component	0.5%	1.5%	1.1%	7.0%

Table 4: The overall effect of individual systematic uncertainties on all backgrounds measured in selections with and without the application of the tau trigger with a p_T threshold of 25 GeV. The systematic uncertainties are shown for 1-track and 3-track $\tau_{\text{had-vis}}$ candidates separately. If the systematic uncertainty consists of both upward and downward variations, the variation resulting in the highest effect is shown. Systematic uncertainties (or pairs thereof) whose overall effect is 0.05% or less are not shown.

6.4 Offline $\tau_{\text{had-vis}}$ in-situ energy scale measurement

This section describes the in-situ measurement of the tau energy scale (TES) based on collision data. This measurement is used to derive an extra correction to account for differences between data and simulation. The method is based on the fact that the distribution of the reconstructed visible mass, m_{vis} , in $Z \rightarrow \tau_\mu \tau_{\text{had}}$ events is sensitive to shifts in the TES between data and simulation.

In this measurement, the event selection is described in section 6.1. The signal and background estimation is the same as that used in the trigger efficiency measurement, as described in section 6.2.

6.4.1 Energy scale extraction

The m_{vis} variable is defined as the invariant mass of the $\tau_{\text{had-vis}}$ and muon system. The tau energy is parametrised as $E_T \rightarrow (1 + \alpha)E_T$ by introducing the TES shift factor α , while the muon momentum scale is measured independently with high precision. In Run 1, α was determined by comparing the m_{vis} fitted peak value between data and simulation [2]. One drawback of this peak-fit method is that the peak value is easily affected by statistical fluctuations. In Run 2, a more robust method has been developed by comparing the full m_{vis} shape and α is determined by finding the value at which the simulation and data maximally agree. Technically, α is determined by minimising $\chi^2(\alpha, f)$, defined as:

$$\chi^2(\alpha, f) = \sum_i \frac{\left(N_i^{\text{data}} - f N_i^{\text{sig}}(\alpha) - N_i^{\text{bkg}}\right)^2}{\left(\sqrt{N_i^{\text{data}}}\right)^2 + f^2 \left(\Delta N_i^{\text{sig}}(\alpha)\right)^2 + \left(\Delta N_i^{\text{bkg}}\right)^2}. \quad (9)$$

Here $N_i^{\text{data/sig/bkg}}$ is the number of events in the i -th bin of the visible mass distribution in the data, signal or background; $\Delta N_i^{\text{sig/bkg}}$ is the corresponding uncertainty in the number of events; N_i^{bkg} (ΔN_i^{bkg}) corresponds to the sum of the contributions from all backgrounds; the parameter f is introduced to reduce the impact of overall normalisation discrepancies between data and simulation. The signal yield in each bin depends upon the in-situ TES parameter α .

6.4.2 Results

For the baseline tau energy calibration, the measured TES shift factor is $\alpha = -0.7\% \pm 0.8\%$ (stat) $\pm 1.2\%$ (syst) and $\alpha = -3.6\% \pm 1.2\%$ (stat) $\pm 3.0\%$ (syst) for $\tau_{\text{had-vis}}$ with one and three associated tracks, respectively. The corrections are applied to the momentum of $\tau_{\text{had-vis}}$ in simulation in order to yield agreement (on average) with data. The resulting m_{vis} distribution for data and simulation is shown in Figure 14 after applying the TES correction. For the BRT tau energy calibration, the measured TES shift factor is $\alpha = 0.95\% \pm 0.9\%$ (stat) $\pm 1.7\%$ (syst) and $\alpha = -3.1\% \pm 1.1\%$ (stat) $\pm 1.6\%$ (syst) for $\tau_{\text{had-vis}}$ with one and three associated tracks, respectively.

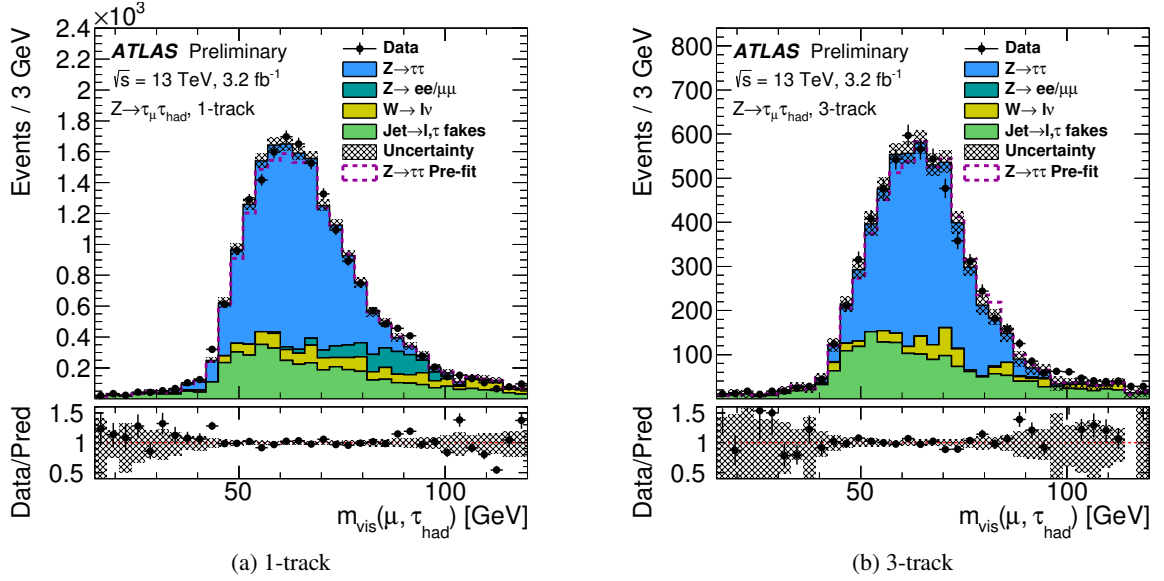


Figure 14: The distribution of m_{vis} : the invariant mass of the $\tau_{\text{had-vis}}$ and muon system. The uncertainty band contains the statistical uncertainty only.

As summarised in Table 5, the dominant systematic uncertainties of the measurement of the baseline TES calibration are due to background modelling, tau energy resolution, and muon performance. The uncertainty due to the background modelling is estimated conservatively by varying the tau identification working points, varying the fit range and bin width, and from the normalisation uncertainties propagated from the uncertainty of the normalisation factors derived from the control regions. The uncertainties due to the muon reconstruction include the uncertainties on the reconstruction, identification and resolution. The systematic uncertainties of the measurement of the BRT TES calibration are estimated in a similar way.

Source	Uncertainty [%]	
	1-track	3-track
Background modelling	1.1	2.9
Tau energy resolution	0.3	0.6
Muon reconstruction	0.2	0.6
Total	1.2	3.0

Table 5: Dominant systematic uncertainties on the tau energy scale correction factor estimated using the in-situ method. The values depend on the number of associated tracks. All other systematic uncertainties are smaller than 0.1%.

7 $t\bar{t}$ tag-and-probe trigger efficiency measurement

The higher mass of the top quark in comparison to the Z boson results in decays to tau leptons with a harder p_T spectrum. This enables performance measurements in the p_T region that is difficult to access and provides a useful cross check to the $Z \rightarrow \tau\tau$ analyses discussed in section 6. This section describes the tau trigger efficiency measurement using the process $t\bar{t} \rightarrow [b\mu\nu_\mu][b\tau\nu_\tau]$, in which the muon constitutes the ‘tag’ object, and the hadronically decaying $\tau_{\text{had-vis}}$ is probed. Signal events are those which contain a $\tau_{\text{had-vis}}$ candidate geometrically matched with a generated $\tau_{\text{had-vis}}$.

7.1 Event selection

The selection and triggering of the muon candidate is the same as described in section 6. Likewise, events with additional electrons or muons are vetoed and at least one opposite sign charge $\tau_{\text{had-vis}}$ is required, with the leading p_T candidate considered. Non- $t\bar{t}$ processes are suppressed by requiring at least two jets with $p_T > 20$ GeV in the event, and with at least one b -tagged.

7.2 Signal and background processes

All simulated events originating from $t\bar{t}$, single top, and W/Z +jets processes where the probe is geometrically matched to a generated, hadronically decaying $\tau_{\text{had-vis}}$ particle are considered as signal events. The main backgrounds are events where a quark- or gluon-initiated jet is reconstructed and misidentified as a $\tau_{\text{had-vis}}$ object. These backgrounds principally result from multi-jet processes as well as $t\bar{t}$, single top, and W/Z +jets processes. In similarity to section 6.3.1, as described in equation 6 and 7, the combined shape of these backgrounds is taken from events in which the muon and the $\tau_{\text{had-vis}}$ have the same sign charge. Normalisation factors (r_{QCD}) for the backgrounds are derived in a control region enriched in multi-jet events, defined by inverting the isolation requirement on the muon and dropping the b -tag requirement. The normalisation factors are defined as the ratio of OS to SS charge events and are parametrised by the number of tracks associated with the $\tau_{\text{had-vis}}$ candidate, as well as its p_T . The r_{QCD} value is computed separately for events before and after the application of the tau trigger.

In similarity to section 6.3.1, simulated events are used to model the excess of the opposite sign events to same sign events for events containing $t\bar{t}$, single top, and W/Z +jets processes, where a jet is misidentified

as the $\tau_{\text{had-vis}}$. The small background from events in which an electron or muon is misidentified as the probe is also modelled by simulated events with the subset of events with the same sign requirement subtracted. Figure 15 shows the signal region tau p_T distributions before and after the application of the tau trigger.

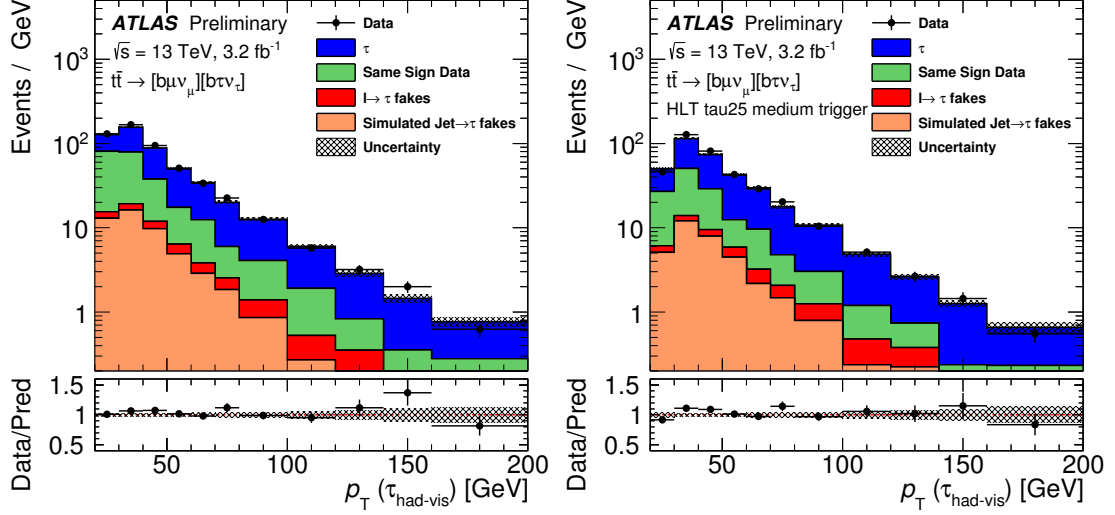


Figure 15: Tau candidate p_T distribution after the $t\bar{t}$ selection, before (left) and after (right) applying the tau trigger with threshold $p_T > 25$ GeV. Here, ‘Simulated Jet $\rightarrow \tau$ fakes’ represents the simulated jets misidentified as tau candidates from $t\bar{t}$, single top, and W/Z +jets processes with opposite sign charged events subtracted by same sign charged events.

7.3 Results

The tau trigger efficiency is measured with respect to tau candidates reconstructed and identified offline in the same manner as described in section 6.3, and as a function of the reconstructed transverse momentum of the tau candidate. For the tau trigger with a p_T threshold of 25 GeV, described in 6.3, and for events with tau candidates reconstructed with the offline medium identification requirement, the efficiencies and corresponding scale factors in simulated signal and data events (with the estimated backgrounds subtracted) are shown in Figure 16. The efficiencies agree above 40 GeV for 1-track $\tau_{\text{had-vis}}$ candidates, and above 45 GeV for 3-track $\tau_{\text{had-vis}}$ candidates.

Sources of systematic uncertainty include the reconstruction and identification efficiencies and the energy scale of the muon, the reconstruction efficiency of the b -jets, and the estimation of the r_{QCD} normalisation factor. With the exception of r_{QCD} , all are correlated with and without the tau trigger applied. The systematic uncertainties enter through the subtraction of estimated backgrounds from data. The main sources of systematic uncertainty are displayed in Table 6. The largest source of uncertainty, the multi-jet normalisation factor, contains a statistical component related to the number of events in the SS control region, and a systematic component derived by varying the choice of selection criteria used to define the control region, and so covers the uncertainty in the assumption that the OS to SS ratio is the same between the isolation and anti-isolation regions. With respect to the $Z \rightarrow \tau\tau$ analysis in Section 6.3, the $t\bar{t}$ analysis improves the precision of the trigger efficiency measurement for tau candidates with $p_T > 100$ GeV.

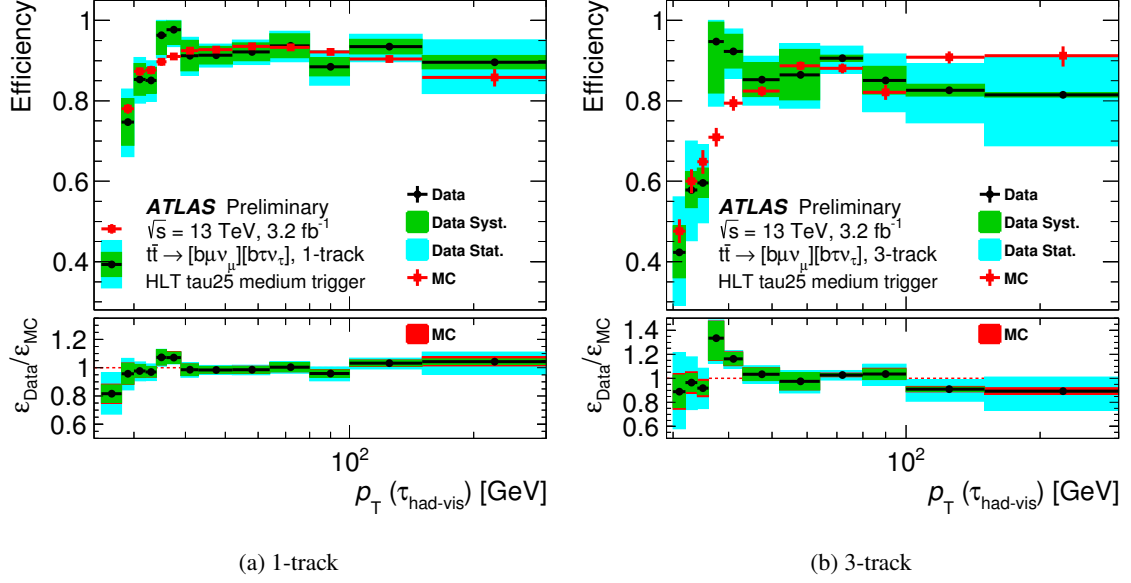


Figure 16: Efficiencies for signal and background subtracted data as well as the corresponding scale factors ($\epsilon_{\text{Data}}/\epsilon_{\text{MC}}$) for the tau trigger with online $p_T > 25$ GeV, as a function of the transverse momentum of offline $\tau_{\text{had-vis}}$ candidates with a medium identification requirement. “Data Syst.” (“Data Stat.”) reflects the systematic (statistical) uncertainty associated with data; “MC” includes only the statistical uncertainty of the simulation. The spikes in the efficiency from data in the above two plots are due to the statistical fluctuation in the background subtraction.

Source	Uncertainty [%]	
	1-track	3-track
multi-jet normalisation	7.7	13.7
b -tagging	< 1	< 1
muon	< 1	< 1

Table 6: Dominant systematic uncertainties on the estimated backgrounds for the $t\bar{t}$ tag-and-probe trigger efficiency measurement.

8 $Z \rightarrow ee$ tag-and-probe analysis

The probability of misidentifying an electron as a tau candidate is measured in events dominated by $Z \rightarrow ee$ processes. As in the previous sections, a tag-and-probe approach is used. Events are selected by triggering on the presence of an electron (tag) and must contain a tau candidate decaying hadronically with a single track (probe). In this section, signal refers to $Z \rightarrow ee$ events containing a $\tau_{\text{had-vis}}$ candidate geometrically matched with a generated electron.

8.1 Event selection

The event selection is chosen to produce a sample of events enriched in $Z \rightarrow ee$ processes. The tag object in the event is a reconstructed electron with $p_T > 25$ GeV, and *tight* likelihood identification requirements. Several triggers are available, with online medium electron likelihood identification and varying online p_T requirements. These select events containing electrons with varying levels of efficiency, and therefore the trigger used to select events is dependent on the p_T of the reconstructed electron in order to maximise acceptance. Events in which the reconstructed electron has offline $p_T > 135$ GeV, $65 \text{ GeV} < p_T < 135 \text{ GeV}$ or $25 < p_T < 65 \text{ GeV}$, are selected and geometrically matched to a trigger object with a respective online requirement of $p_T > 120$ GeV, $p_T > 60$ GeV or $p_T > 24$ GeV. The probe $\tau_{\text{had-vis}}$ candidate must have a single track, $p_T > 20$ GeV, and a veto is placed on events containing muons or *b*-tagged jets.

To ensure the selected events contain a high purity of $Z \rightarrow ee$ decays, an additional selection is placed on the signal region, requiring the electron tag $p_T > 30$ GeV, the invariant mass of the electron and tau system to be within $80 \text{ GeV} < m_{\text{vis}}(e, \tau_{\text{had-vis}}) < 100 \text{ GeV}$, and the transverse mass of the electron and E_T^{miss} system, $m_T(e, E_T^{\text{miss}})$, to be less than 40 GeV.

8.2 Signal and background processes

The data sample enriched in $Z \rightarrow ee$ processes is compared to an estimation of signal events and background processes. The $Z \rightarrow ee$ signal process is estimated via simulation and the probe tau from both signal and background processes must be within a cone of size $\Delta R < 0.2$ of a generated electron for simulated events. The contribution to the signal region from $Z \rightarrow \tau\tau$ and top-quark processes are also estimated from simulation.

After the final selection, the fraction of $Z \rightarrow ee$ events is approximately 98%, as shown in Figure 17. W +jets processes are estimated via simulated events, with a normalisation factor extracted from a W +jets dominated control region to correct the imperfect nature of the simulation. The control region is defined by requiring exactly one electron in the event, $m_{\text{vis}}(e, \tau_{\text{had-vis}}) < 80 \text{ GeV}$, $m_T(e, E_T^{\text{miss}}) > 70 \text{ GeV}$, and $E_T^{\text{miss}} > 30 \text{ GeV}$.

The shape of the multi-jet background is taken from events in which the electron and the tau lepton have the same charge and scaled with a normalisation factor derived in a control region enriched in multi-jet events. The multi-jet control region is defined by inverting the isolation requirement on the electron, such that the ratio of transverse momentum in a cone of $\Delta R < 0.2$ around the electron to the electron p_T , and the ratio of transverse momentum in a cone of $\Delta R < 0.4$ around the electron to the electron p_T , are

greater than 12% and 8% respectively. Same sign events from the simulated $Z \rightarrow \tau\tau$, W +jets and top backgrounds are subtracted from the same sign background.

The p_T and η distributions of the tau candidate are shown in Figure 17 after the full event selection and with the background estimation described above.

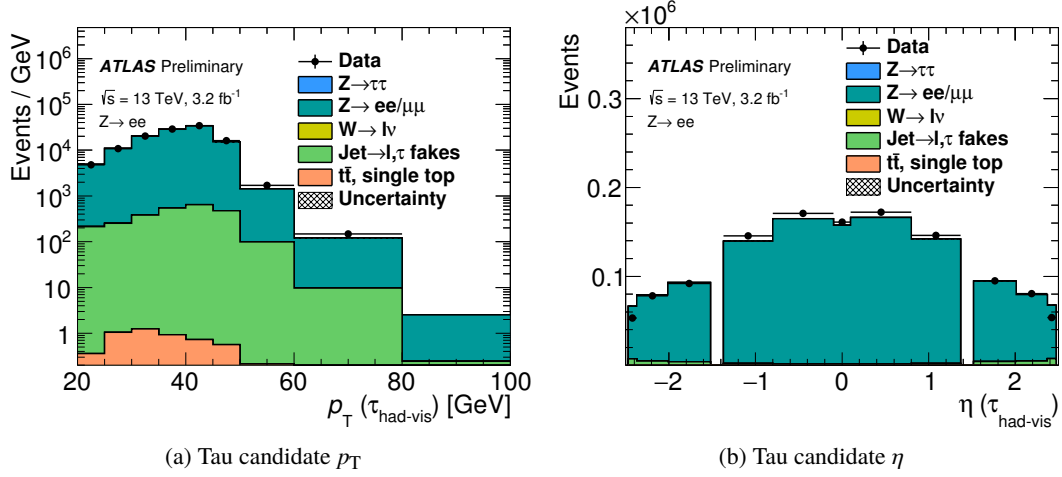


Figure 17: The p_T and η distributions of the tau candidate after the full $Z \rightarrow ee$ event selection. In Figure (b), the yield in each bin is normalized according the bin width.

8.3 Results

The electron misidentification probability is defined as the probability of an electron satisfying both the tau identification and the electron discrimination algorithms. It is measured by taking the ratio of signal region events passing the medium tau identification and the *very loose* electron discrimination requirements to all signal region events, and is calculated for both data and the simulated $Z \rightarrow ee$ signal process. For the efficiency in data, the estimated contribution to the signal region coming from non- $Z \rightarrow ee$ events is first subtracted. The electron misidentification probability ranges between 0.5% and 2.5% across the η spectrum of the tau candidate in both data and simulation. The correction factors between the efficiencies in data and simulation are shown in Figure 18. The data to simulation agreement is constant across the $|\eta(\tau_{\text{had-vis}})|$ range, except for $|\eta(\tau_{\text{had-vis}})| < 0.1$ and $|\eta(\tau_{\text{had-vis}})| > 2.4$, where boundaries of the detector coverage are present.

Sources of uncertainty are shown in Table 7. Systematic uncertainties associated with the reconstruction, identification and energy scale of the electron, and the electron trigger are considered. The estimation of the multi-jet and W +jets normalisation factors contribute as additional sources of uncertainty, and include a statistical component as well as a systematic component of 10%, conservatively estimated from the $Z \rightarrow \tau_\mu \tau_{\text{had}}$ measurements. The uncertainty in the normalisation factors covers the assumption that the OS to SS ratio is the same between the signal and control regions. The uncertainties are found consistent between different tau identification working points.

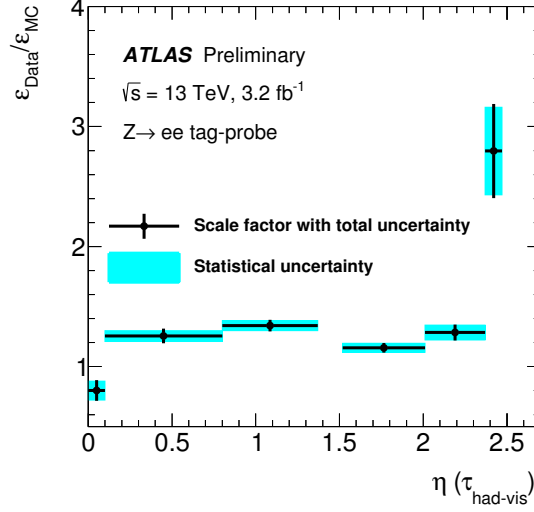


Figure 18: Electron misidentification correction scale factors between data and simulation for the requirement that $\tau_{\text{had-vis}}$ candidates with overlapping electrons pass a medium tau identification criterion and the electron discrimination algorithm. The measurements are carried out on data enriched in $Z \rightarrow ee$ events and with estimated backgrounds subtracted, as well as simulated $Z \rightarrow ee$ events, and the ratio ($\epsilon_{\text{Data}}/\epsilon_{\text{MC}}$) is displayed as a scale factor.

Source	Uncertainty [%]
Statistics	4.9
Multi-jet normalisation factor	< 1
W +jets scale factor	< 1
Electron	< 1

Table 7: Dominant uncertainties on the electron misidentification scale factors estimated with the Z boson tag-and-probe method.

9 Summary and conclusions

Table 8 summarises the tau performance measurements and their usage in physics analyses. The performance of the online and offline tau identification algorithms, and the energy calibration are measured using $Z \rightarrow \tau\tau$ tag-and-probe measurements. The uncertainties on the offline tau identification efficiency correction factor measurement are approximately 5% (6%) for one (three) track, inclusive in η and for a visible transverse momentum greater than 20 GeV. The online tau identification efficiency correction factor is measured with a precision of (3–20)% depending on the transverse energy of the tau candidate selected by offline algorithms. The online tau identification efficiency correction factor measurement from $t\bar{t}$ processes is complementary in the higher p_T range with a precision of (8–14)%. The probability of misidentifying electrons as tau candidates is measured to be < 2.5%, and the corresponding simulation to data correction factors are measured with a precision of 3–14%. The reconstructed tau energy scale correction factor is measured with a precision of approximately (2–3)%.

Tau performance	Measurements	Precision (%)
Offline identification	$Z \rightarrow \tau\tau$	5–6
Energy scale	$Z \rightarrow \tau\tau$	2–3
Trigger	$Z \rightarrow \tau\tau$	3–20
	$t\bar{t} \rightarrow [b\mu\nu_\mu][b\tau\nu_\tau]$	8–14
Electron veto	$Z \rightarrow ee$	3–14

Table 8: Summary of the tau performance measurements and the relative precision on the simulation to data correction factor. The correction factor of the above measurements are all applied to the generated tau candidates in physics analyses, except “Electron veto” which is applied to the simulated tau candidates misidentified from electrons. For the trigger efficiency scale factor, the measurement from $Z \rightarrow \tau\tau$ is used for the p_T threshold of 25, 35, 50 GeV, while the measurement from $t\bar{t} \rightarrow [b\mu\nu_\mu][b\tau\nu_\tau]$ is used for the p_T threshold of 80, 125, 160 GeV.

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Appendix

A Tau Identification input variables of $Z \rightarrow \tau_\mu \tau_{\text{had}}$ tag-and-probe analysis

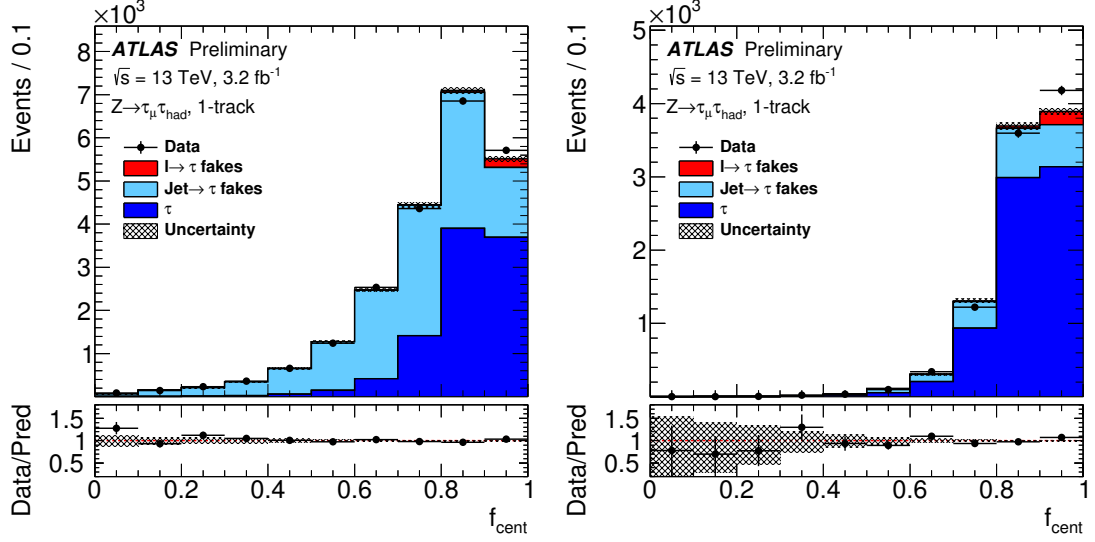


Figure 19: Central energy fraction f_{cent} , as defined in Figure 9, for one track $\tau_{\text{had-vis}}$ candidates before (left) and after (right) medium identification requirement. The uncertainty band includes only the statistical uncertainty.

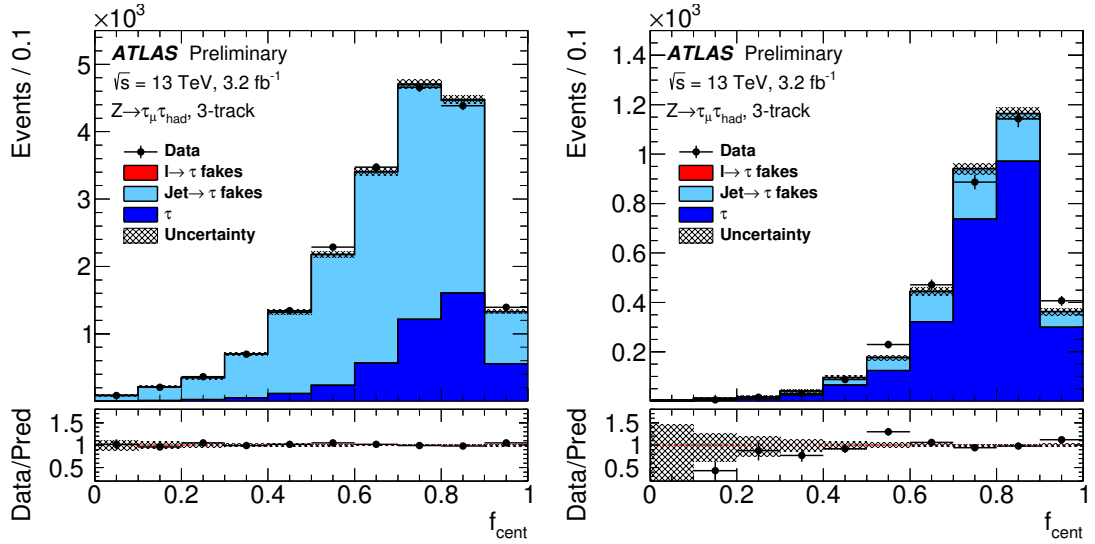


Figure 20: Central energy fraction f_{cent} , as defined in Figure 9, for three track $\tau_{\text{had-vis}}$ candidates before (left) and after (right) medium identification requirement. The uncertainty band includes only the statistical uncertainty.

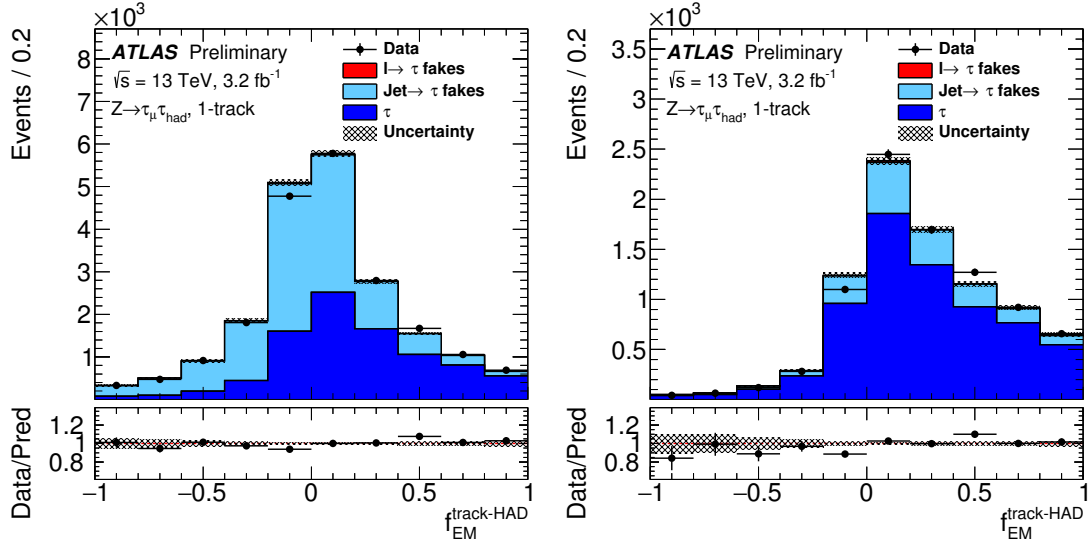


Figure 21: Fraction of EM energy from charged pions $f_{\text{EM}}^{\text{track-HAD}}$, as defined in Figure 7, for one track $\tau_{\text{had-vis}}$ candidates before (left) and after (right) medium identification requirement. The uncertainty band includes only the statistical uncertainty.

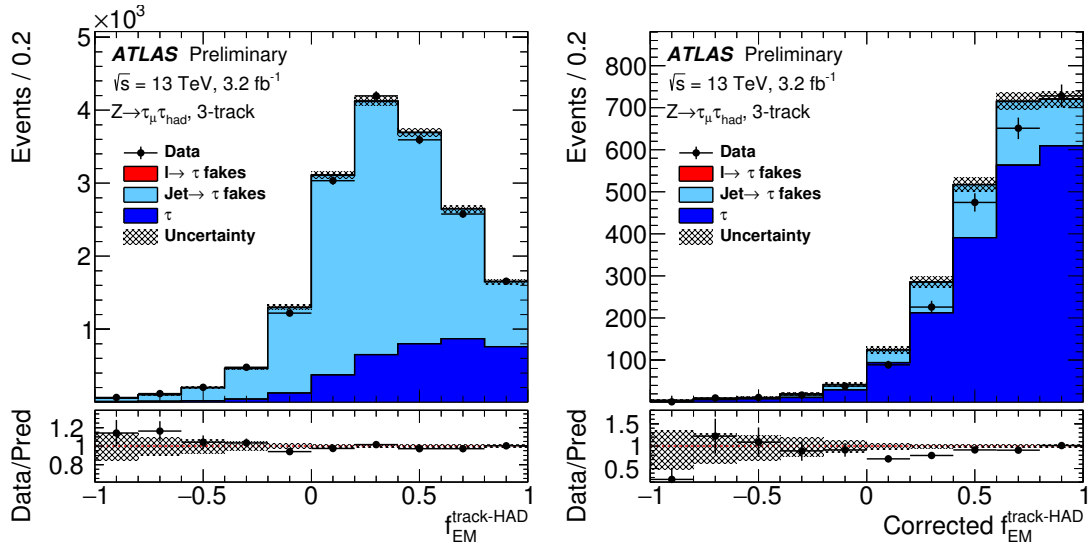


Figure 22: Fraction of EM energy from charged pions $f_{\text{EM}}^{\text{track-HAD}}$, as defined in Figure 7, for three track $\tau_{\text{had-vis}}$ candidates before (left) and after (right) medium identification requirement. The uncertainty band includes only the statistical uncertainty.

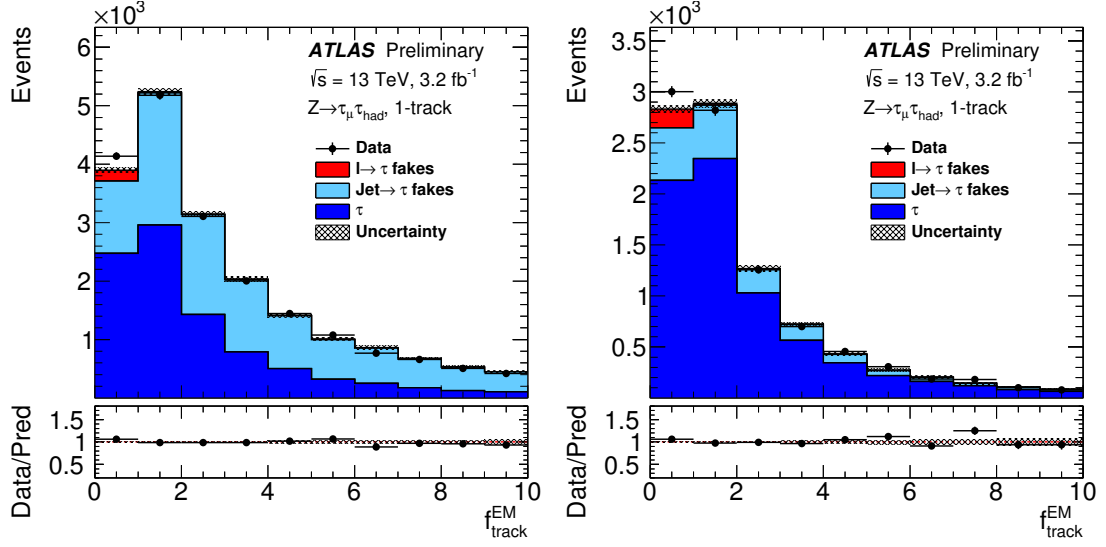


Figure 23: Ratio of EM energy to track momentum $f_{\text{track}}^{\text{EM}}$ for one track $\tau_{\text{had-vis}}$ candidates before (left) and after (right) medium identification requirement. $f_{\text{track}}^{\text{EM}}$ is defined as the ratio of the sum of energy deposited in the electromagnetic calorimeter associated with the $\tau_{\text{had-vis}}$ candidate to the sum of the momentum of tracks in the core region. The uncertainty band includes only the statistical uncertainty.

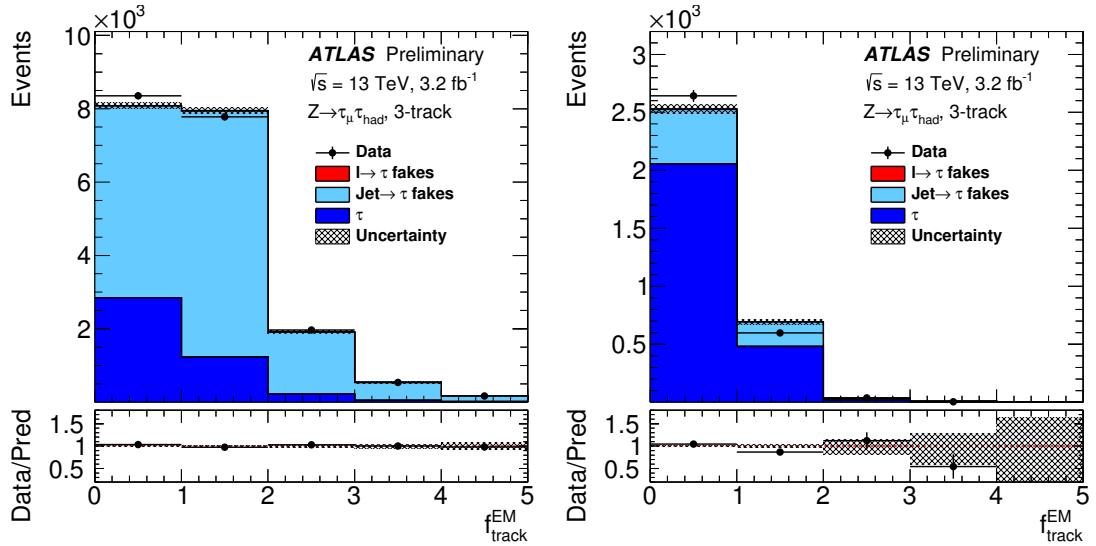


Figure 24: Ratio of EM energy to track momentum $f_{\text{track}}^{\text{EM}}$, as defined in Figure 23, for three track $\tau_{\text{had-vis}}$ candidates before (left) and after (right) medium identification requirement. The uncertainty band includes only the statistical uncertainty.

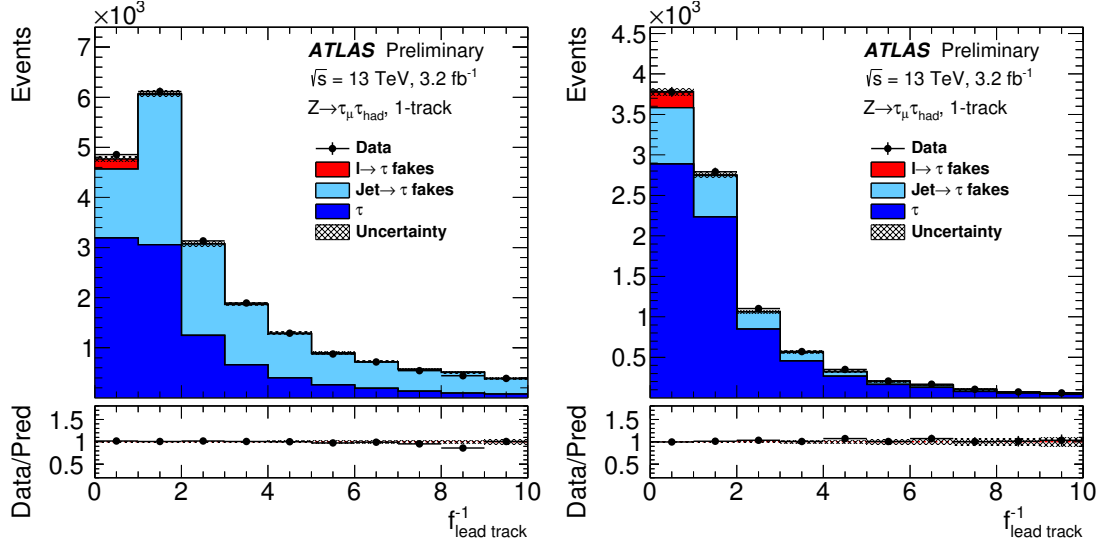


Figure 25: Leading track momentum fraction $f_{\text{lead track}}^{-1}$ for one track $\tau_{\text{had-vis}}$ candidates before (left) and after (right) medium identification requirement. $f_{\text{lead track}}^{-1}$ is defined as the transverse energy sum deposited in all cells belonging to TopoClusters in the core region of the $\tau_{\text{had-vis}}$ candidate, divided by the transverse momentum of the highest p_T charged particle in the core region. The uncertainty band includes only the statistical uncertainty.

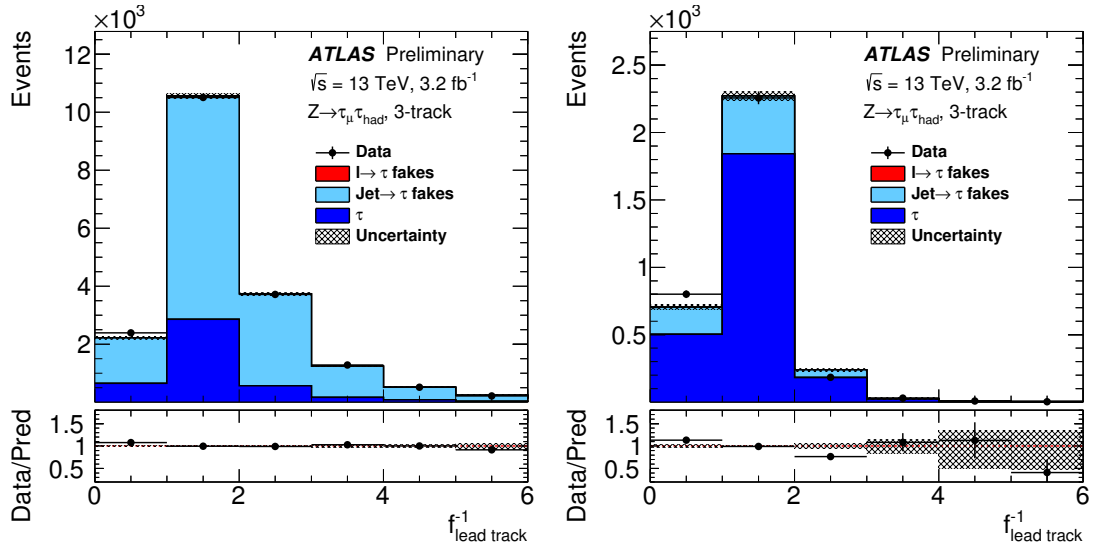


Figure 26: Leading track momentum fraction $f_{\text{lead track}}^{-1}$, as defined in Figure 25, for three track $\tau_{\text{had-vis}}$ candidates before (left) and after (right) medium identification requirement. The uncertainty band includes only the statistical uncertainty.

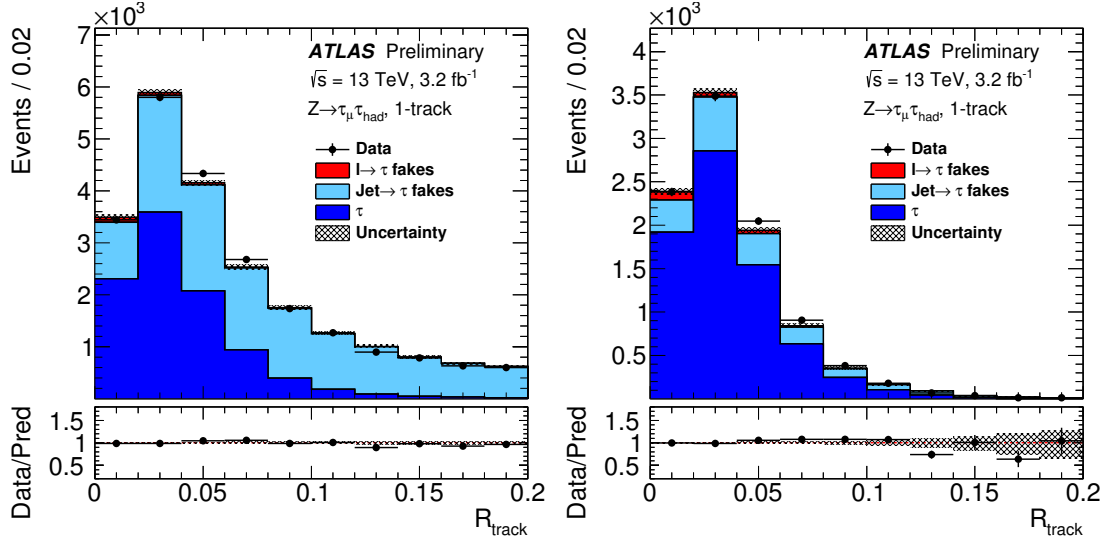


Figure 27: Track radius R_{track} for one track $\tau_{\text{had-vis}}$ candidates before (left) and after (right) medium identification requirement. R_{track} is defined as the p_T -weighted ΔR distance of the associated tracks to the $\tau_{\text{had-vis}}$ direction, using only tracks in the core region. The uncertainty band includes only the statistical uncertainty.

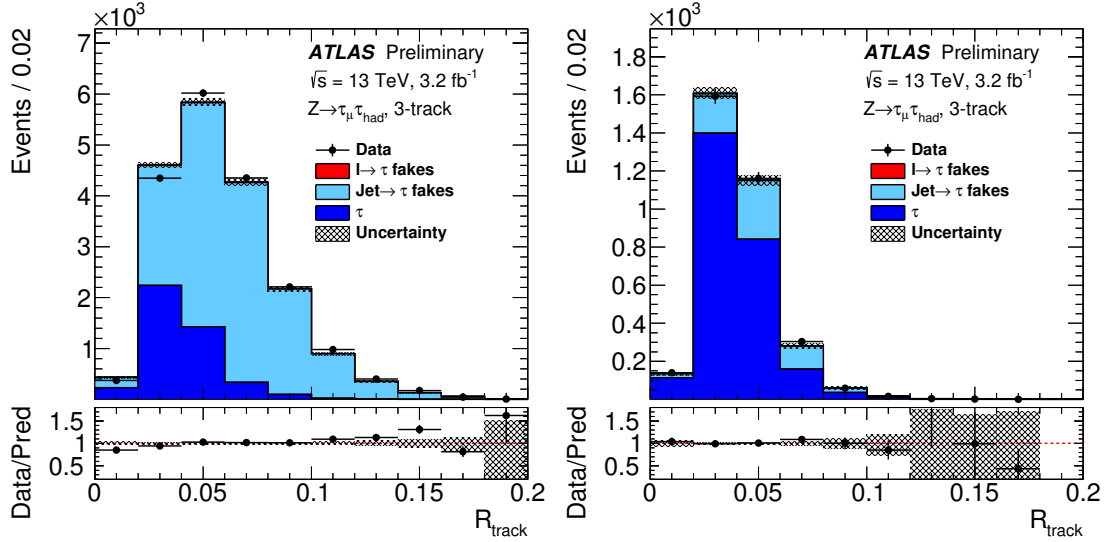


Figure 28: Track radius R_{track} , as defined in Figure 27, for three track $\tau_{\text{had-vis}}$ candidates before (left) and after (right) medium identification requirement. The uncertainty band includes only the statistical uncertainty.

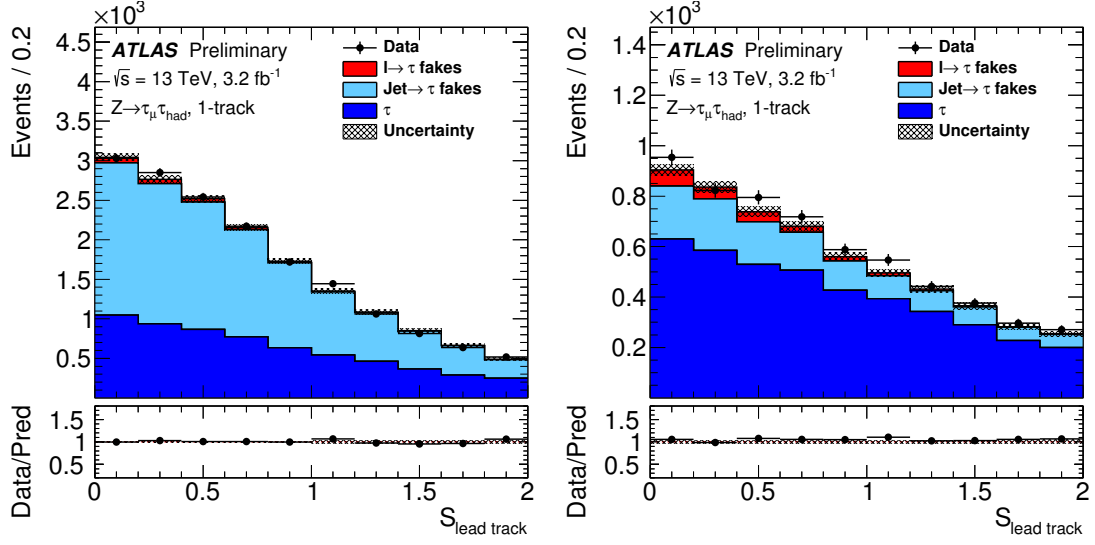


Figure 29: Leading track IP significance $S_{\text{lead track}}$ for one track $\tau_{\text{had-vis}}$ candidates before (left) and after (right) medium identification requirement. $S_{\text{lead track}}$ is defined as the absolute value of transverse impact parameter of the highest p_T track in the core region, calculated with respect to the tau vertex, divided by its estimated uncertainty. The uncertainty band includes only the statistical uncertainty.

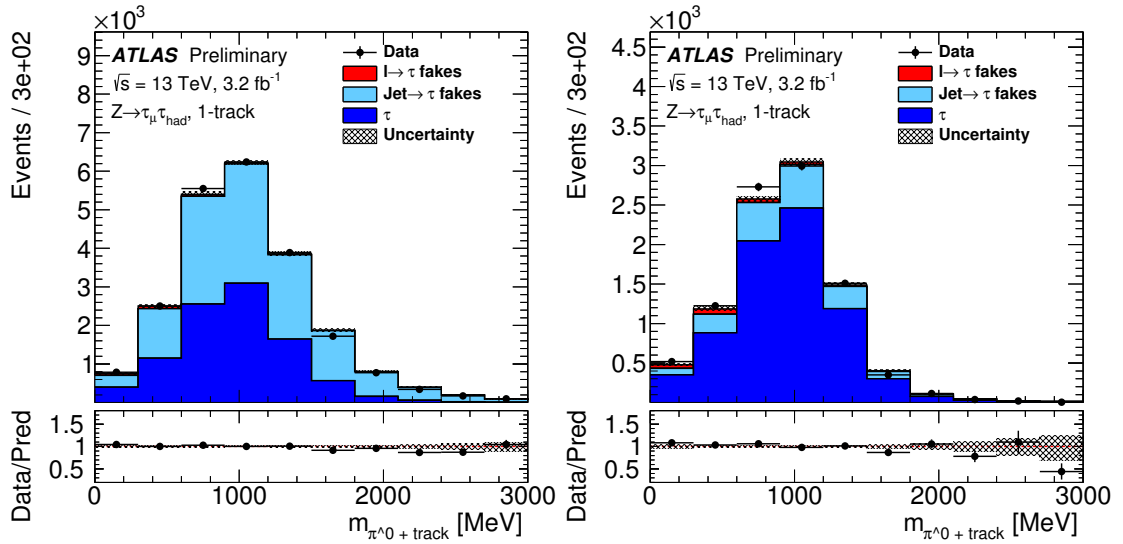


Figure 30: Track-plus-EM-system mass $m_{\pi^0+\text{track}}$ for one track $\tau_{\text{had-vis}}$ candidates before (left) and after (right) medium identification requirement. $m_{\pi^0+\text{track}}$ is defined as the invariant mass of the system composed of the tracks and up to two most energetic electromagnetic calorimeter clusters in the core region. The uncertainty band includes only the statistical uncertainty.

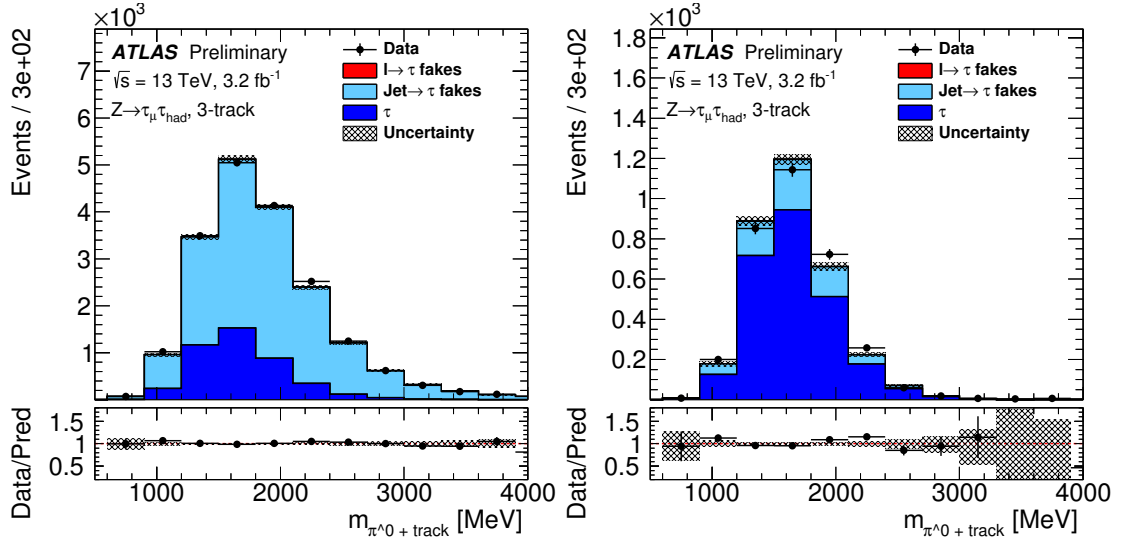


Figure 31: Track-plus-EM-system mass $m_{\pi^0 + \text{track}}$, as defined in Figure 30, for three track $\tau_{\text{had-vis}}$ candidates before (left) and after (right) medium identification requirement. The uncertainty band includes only the statistical uncertainty.

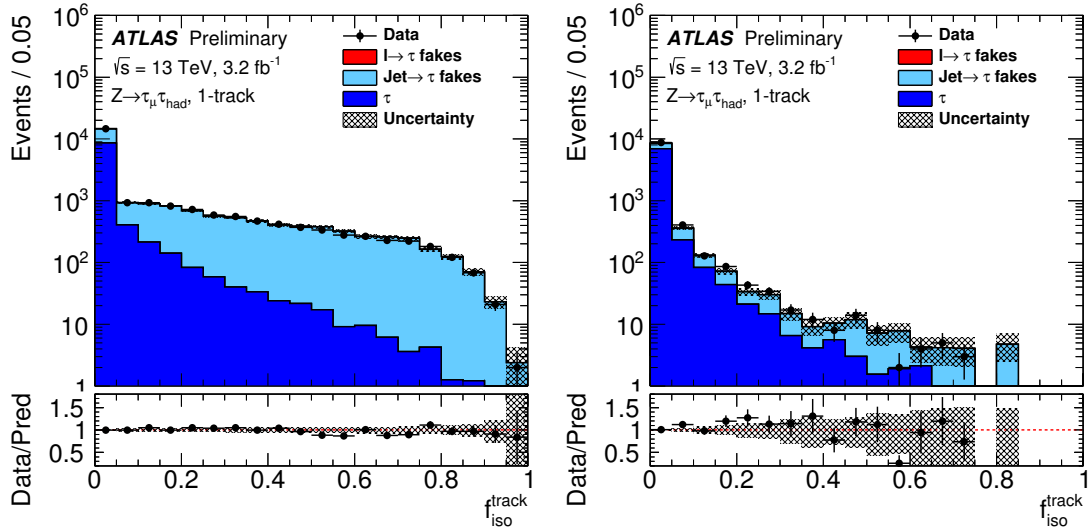


Figure 32: Fraction of tracks p_T in the isolation region $f_{\text{iso}}^{\text{track}}$, as defined in Figure 8, for one track $\tau_{\text{had-vis}}$ candidates before (left) and after (right) medium identification requirement. The uncertainty band includes only the statistical uncertainty.

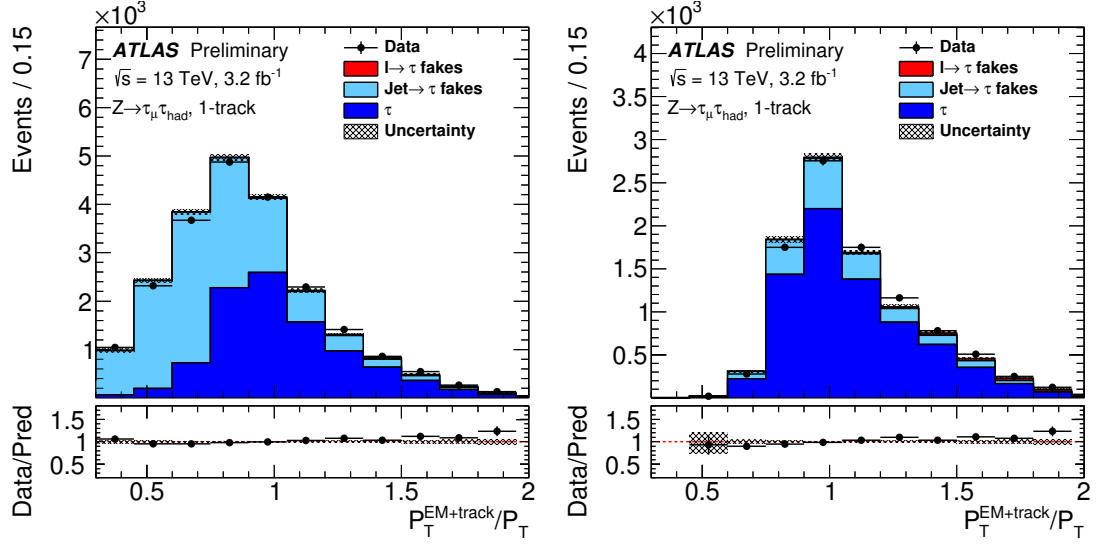


Figure 33: Ratio of track-plus-EM-system to p_T $P_T^{\text{EM+track}}/P_T$ for one track $\tau_{\text{had-vis}}$ candidates before (left) and after (right) medium identification requirement. $P_T^{\text{EM+track}}/P_T$ is defined as the ratio of the $\tau_{\text{had-vis}}$ p_T , estimated using the vector sum of track momenta and up to two most energetic electromagnetic calorimeter clusters in the core region, to the calorimeter-only measurement of $\tau_{\text{had-vis}}$ p_T . The uncertainty band includes only the statistical uncertainty.

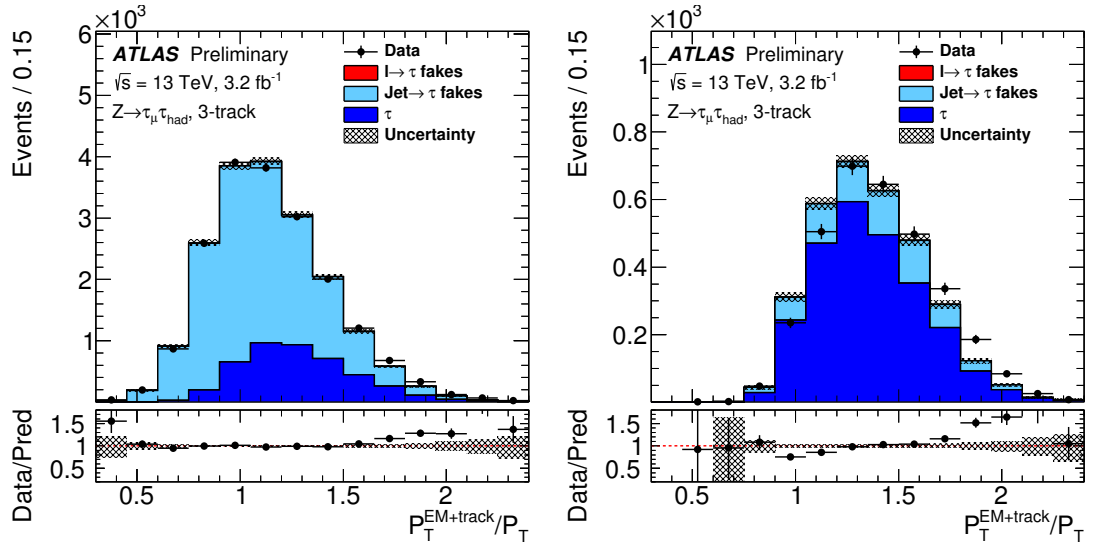


Figure 34: Ratio of track-plus-EM-system to p_T $P_T^{\text{EM+track}}/P_T$, as defined in figure 33, for three track $\tau_{\text{had-vis}}$ candidates before (left) and after (right) medium identification requirement. The uncertainty band includes only the statistical uncertainty.

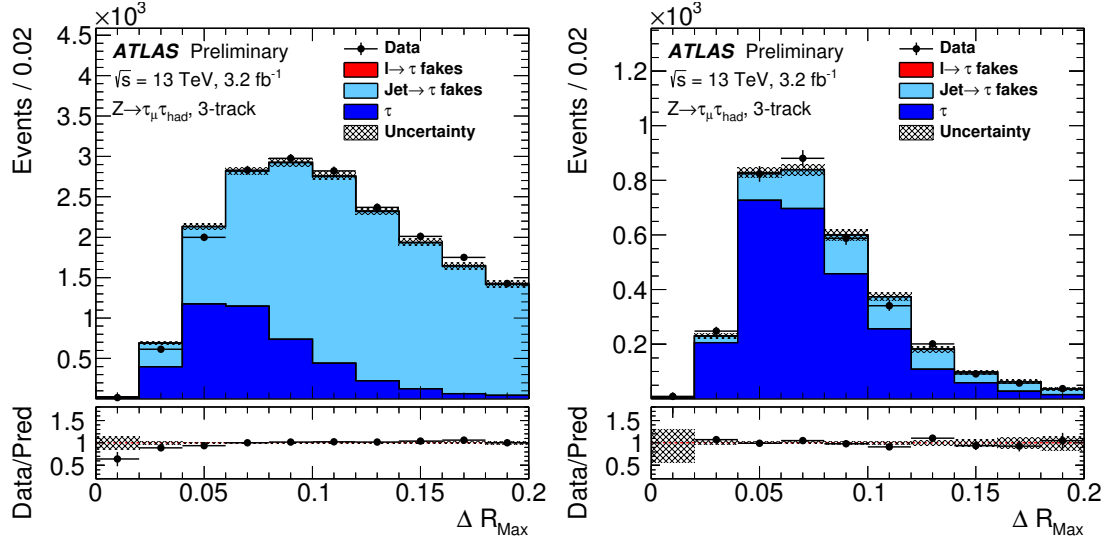


Figure 35: Maximum ΔR_{Max} for three track $\tau_{\text{had-vis}}$ candidates before (left) and after (right) medium identification requirement. ΔR_{Max} is defined as the maximum ΔR between a track associated with the $\tau_{\text{had-vis}}$ candidate and the $\tau_{\text{had-vis}}$ direction. The uncertainty band includes only the statistical uncertainty.

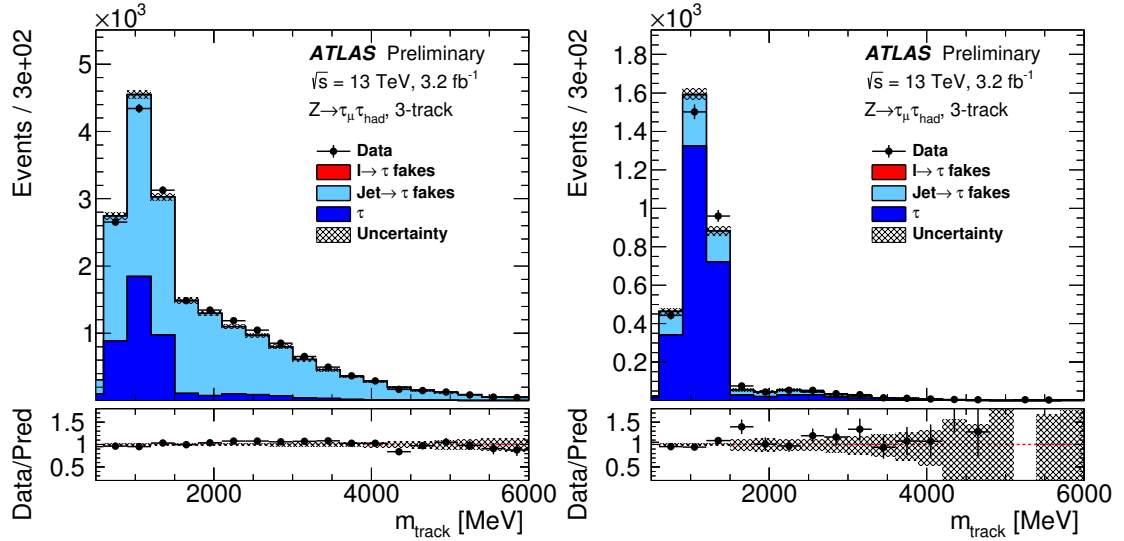


Figure 36: Track mass m_{track} for three track $\tau_{\text{had-vis}}$ candidates before (left) and after (right) medium identification requirement. m_{track} is defined as the invariant mass calculated from the sum of the four-momentum of all tracks in the core and isolation regions, assuming a pion mass for each track. The uncertainty band includes only the statistical uncertainty.

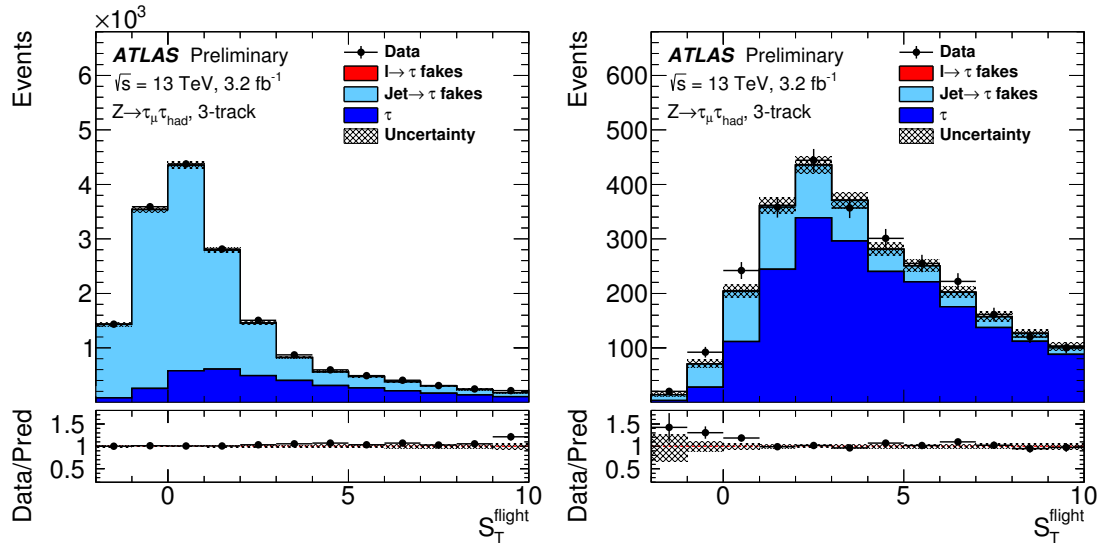


Figure 37: Transverse flight path significance S_T^{flight} for three track $\tau_{\text{had-vis}}$ candidates before (left) and after (right) medium identification requirement. S_T^{flight} is defined as the decay length of the secondary vertex, reconstructed from the tracks associated with the core region of the $\tau_{\text{had-vis}}$ candidate, in the transverse plane, calculated with respect to the tau vertex, divided by the estimated uncertainty. The uncertainty band includes only the statistical uncertainty.